

Program overview

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Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
BME Master 2023			
Master Biomedical Engineering 2023			
BME Master 1st Year			
BME Obligatory courses (11 ECTS)			
BM41141	Fundamentals of Biomedical Engineering	5	
BM51007	Professional Development Journey	1	
WM1402TU	Ethics of Healthcare Technologies	5	
BME Obligatory Electives: select 1 out of 2, depending on your background (4 ECTS)			
BM41055	Anatomy and Physiology	4	
ME41096	Bio Inspired Design	5	
BME Elective courses for all tracks			
BM41070	Medical Device Prototyping	6	
BM41110	Clinical Patient Safety	2	
BME-NMB Track NEUROMUSCULOSKELETAL BIOMECHANICS (NMB)			
BME-NMB Obligatory (25 ECTS)			
BM41040	Neuromechanics & Motor Control	5	
BM41090	Computational Mechanics of Tissues and Cells	6	
BM41131	Tissue Biomechanics	3	
ME41065	System Identification and Parameter Estimation	7	
ME41085	Biomechatronics	4	
BME-NMB Obligatory electives: Select 3 courses from the following list			
BM41035	Biomaterials	4	
BM41155	3D Printing	4	
BM41170	Physiological Flows and Transport	4	
ME41006	Musculoskeletal Modeling and Simulation	4	
ME41035	Special Topics in Sports Engineering	3	
ME41056	Multibody Dynamics	5	
RO47006	Human Robot Interaction	5	
SC42001	Control System Design	5	
SC42015	Control Theory	6	
BME-MD Track MEDICAL DEVICES (MD)			
BME-MD Obligatory courses (24 ECTS)			
BM41035	Biomaterials	4	
BM41050	Applied Experimental Methods: Medical Instruments	4	
BM41095	Medical Devices: Past, Present, Future	3	
BM41155	3D Printing	4	
EE4555	Active Implantable Biomedical Microsystems	5	
ET4127	Themes in Biomedical Electronics	4	
BME-MD Obligatory electives: Select 3 courses from the following list			
BM41075	Regenerative Medicine	4	
BM41090	Computational Mechanics of Tissues and Cells	6	
BM41100	Medical Devices: Designing for CE-marking	3	
BM41131	Tissue Biomechanics	3	
EE4109	Structured Electronic Design	5	
EE4C08	Measurement and Instrumentation	5	
EE4C11	Systems Engineering	5	
ET4130	Bioelectricity	3	
ET4257	Sensors and Actuators	4	
ET4386	Estimation and Detection	5	
ME41065	System Identification and Parameter Estimation	7	
BME-MP Track Medical Physics (MP)			
BME-MP Track MEDICAL PHYSICS (MP)			
BME-MP Obligatory courses (24 ECTS)			
AP3132	Advanced Digital Image Processing	6	
AP3232	Medical Imaging Signals and Systems	6	
AP3371	Radiological Health Physics	6	
AP3582	Medical Physics of Photon and Proton Therapy	6	
BME-MP Obligatory electives: Select at least 10 ECTS from the following list			
AP3531	Acoustical Imaging	6	
CS4240	Deep Learning	5	
IN4307	Medical Visualization	5	
NB4020	High-Resolution Imaging	4	

BME Master 2nd Year			
BME Project / Electives - Select 1 out of 3: Internship, JIP or Elective Courses (15 ECTS)			
BM51015	BME Research Internship	15	
TUD4040	Joint Interdisciplinary Project	15	
BME Graduation Project Obligatory (45 ECTS)			
BM51010	BME Literature Research	10	
BM51035	BME MSc Thesis	35	

Year	2023/2024
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BME Master 2023

MSc Coordinator	A.J. Loeve
Program Coordinator	Biomedical Engineering is an interdisciplinary field that integrates the advances of engineering sciences, (bio)medical sciences and clinical practices to improve human health. Biomedical Engineering problems usually require knowledge and expertise in the following areas: - General knowledge of anatomy, physiology, biology and pathology - Fundamental knowledge in specific areas of engineering and/or physical sciences - Specialist knowledge in how biology/medicine principles are combined with engineering/physical science to tackle specific problems within Biomedical Engineering - Scientific methodology and medical ethics As academically trained professionals, biomedical engineers should understand and deploy soundly designed experiments, use proper statistics and implement good practice in data acquisition, archiving and interpretation as well as patient safety. They also need to possess at least a basic understanding of the main principles of evidence-based medicine. The combination of all of the above-mentioned qualities enables biomedical engineers to impact healthcare quality through development and clinical translation of novel medical technologies. The BME Master program at the TU Delft is a two-year master program and it is based on a long history of teaching and research in biomedical engineering within the three collaborating faculties: - Faculty of Mechanical, Maritime, and Materials Engineering (3mE), Department of Biomechanical Engineering - Faculty of Applied Sciences (AS), Departments of Imaging Physics and of Radiation Science & Technology - Faculty of Electrical Engineering, Mathematics and Computer Science (EEMCS), Department of Microelectronics The Faculty of 3mE (Department of Biomechanical Engineering) coordinates the program. In combining the education and research programs of these three faculties, a broad BME program could be realized. Additionally, the program includes close and intensive collaboration with clinical partners at the Leiden University Medical Centre (LUMC), the Erasmus Medical Centre Rotterdam (Erasmus MC), the Academic Medical Centre Amsterdam (AMC), the Free University Medical Centre (VUMC), the University Medical Center of Utrecht (UMCU) and the Maastricht University Medical Center (MUMC). Clinical partners participate in first-year MSc teaching, and in the tutoring of MSc projects in the second year.
Prerequisites	As of 2018-2019, the MSc BME programme consists of three tracks: - Track I: Neuromusculoskeletal Biomechanics - Track II: Medical Devices - Track III: Medical Physics In general, Track I requires a background in Mechanical Engineering, Track II needs a background in Mechanical Engineering or Electrical Engineering and Track III requires a background in (Applied) Physics. This means that BSc graduates in Mechanical Engineering, Applied Physics or Electrical Engineering from a University of Technology may enter the BME program without any restrictions. Academic BSc graduates holding other degrees may be required to follow a bridging (Pre-master) program depending on their area of study. Graduates from a Dutch polytechnic school (Technische Hogeschool) may also enter the program upon completion of some additional courses (the Pre-Master program) only if they have a BSc that is listed in the PDF "Bridging programme" provided on our Master's webpage here: https://www.tudelft.nl/onderwijs/opleidingen/masters/biomedical-engineering/msc-biomedical-engineering/admission-and-application. Bridging courses generally are in Dutch, so you will need to sufficiently master the Dutch language for these courses. International applicants with a university BSc degree need to follow the TUD admission and application process as outlined on the following website: https://www.tudelft.nl/onderwijs/toelating-en-aanmelding/msc-international-diploma/. For the BME Master, the following university BSc degrees are eligible for admission: Mechanical Engineering, Electrical Engineering, Physics, Applied Physics, Biomedical Engineering.
Introduction 1	The BME master program is divided into three tracks: - Track I: Neuromusculoskeletal Biomechanics (NMB) - Track II: Medical Devices (MD) - Track III: Medical Physics (MP) At the beginning of the study program students must choose their track. Switching between tracks is possible, but students should consider the obligatory courses and additional courses required for each track.
Introduction 2	MSc first year (60EC) The first year of the program consists of: A. Programme-wide courses: --a. Programme-obligatory courses (11 EC) obligatory for everyone in the BME master --b. Programme-obligatory electives (4/5 EC) everyone in the BME master should choose one B. Track-specific courses: --a. Track-obligatory courses (24-25 EC) these are all obligatory in that track --b. Track-obligatory electives (ca 10 EC) here a selection must be made C. Free elective courses (ca 10 EC) Following all programme-obligatory courses (11 EC) and one of the programme-obligatory electives (4 or 5 EC) is obligatory for all students of all tracks. These courses are identical for all tracks on their respective versions of Form 2, the Individual Student Programme form. From the programme-obligatory electives, students are not allowed to follow the course BM41055: Anatomy & Physiology if they have a background in: - Biomedical Engineering/Technology, - Biomedical Sciences, - Clinical Technology, - Human Movements Sciences, - Medical Natural Sciences or - Medicine These students have to choose the "ME41096: Bio-inspired design" course instead. Students always have to follow all the track-specific obligatory courses of the track that they chose (24-25 EC). Depending on the chosen track, several track-specific obligatory electives must be added. Further details are provided in the "Study Guide" document provided on the Brightspace page "Master Biomedical Engineering". The general goal of the courses in this programme is to provide students a solid understanding of fundamentals of the relevant scientific topics as well as the state-of-the-art aspects of biomedical engineering that will allow them to undertake current and future challenges in this multidisciplinary field successfully. The obligatory and elective courses are a combination of both biomedical and engineering courses. Biomedical courses are taught by engineers and clinicians. Clinicians discuss clinical issues and explain their viewpoints, while also covering progress in clinically-related research. Several medical courses can be taken within the educational program of two of our clinical partner

universities, LUMC and VUMC. The students may list these medical courses to a maximum of 10 EC in their study program. From the engineering viewpoint, the emphasis is placed on technical and biophysical aspects, such as the latest advances in design, modeling and simulation, all the time relating this to the engineering background of the students.

All newly enrolled students must register their programme with selected courses using a prescribed template (Form 2), called the Individual Study Program (ISP):

Download the Form 2 of your track from the BME Brightspace (<https://brightspace.tudelft.nl>) page Master Biomedical Engineering or from the 3mE MSc forms website.

Create your individual study programme by selecting courses according to the regulations that you find on the form.

Please check the current study guide and the digital study guide (<https://studiegids.tudelft.nl/>) to ensure that your individual study programme meets the requirements.

Next, check your calendar for conflicting lecture times, and spread your study load evenly over the year.

After that, consult the track coordinator of the track you chose, to ensure that you have optimally prepared your study programme.

Form 2 needs to be signed by both the track coordinator and the student, and the original signed form shall be sent to SPA-3mE@tudelft.nl.

Finally, a copy of the signed form should be sent to the master coordinator (Coordinator-BME-3ME@tudelft.nl).

This should be done no later than the deadline mentioned in the BME Study guide on the Master Biomedical Engineering Brightspace page!

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MSc second year (60EC)

For the second year, the BME students can choose for 15 EC to undertake either an Internship, or select electives, or a Joint Interdisciplinary Project (JIP). Once the student chooses one of these items, it is recommended to do it in the first semester of the second year. The remainder of the year is taken up with the Literature study and the Master thesis project. The internship and the literature study may be undertaken in any order. The Internship, Literature study and the Master thesis should be carried out on an individual base.

Internship/Joint Interdisciplinary Project (15 EC)

The Internship can be carried out in a university medical center, hospital, company or other research institute. Over the course of their Internship, students undertake a project task defined in consultation with the host institute and their TUD-BME internship supervisor. Before they plan their Internship abroad, students must contact the International Office at 3mE (internationaloffice-3me@tudelft.nl) for further information. The Internship should culminate in a report and a reflection. As it is difficult to evaluate Internship achievements, because conditions vary widely and as the aims of an Internship are fulfilled more on a process level than on a product level, the student does not receive a grade but a pass/fail assessment. Details about the forms, procedures and rules involved with internships are provided on the M-BME Brightspace page.

Instead of undertaking an Internship, BME students can join an interdisciplinary team under the frame of the Joint Interdisciplinary Project (15 EC). Information about JIP can be found on the following website:

<https://www.jointinterdisciplinaryproject.nl>.

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Literature study (10 EC)

It is required that students do their Literature study in the same research field as their Master thesis project. The Literature study will result in a report written in journal paper style. The students receive a grade for their literature report that will be uploaded in Osiris by their TUD-BME supervisor.

Master thesis project (35 EC)

The Master thesis project is the final part of the BME programme. Ideally, the project is preferably undertaken in collaboration with a clinical partner (e.g., LUMC, Erasmus MC, VUMC, UMCU, AMC or MUMC). The student, together with the TUD-BME thesis advisor and the external party advisor, will define the research project. The MSc thesis layout should be discussed with the TUD-BME thesis advisor. The students either prepare their MSc thesis as a journal paper with appendices covering detailed information, or write it as a report with a journal paper in the appendix.

The thesis work is evaluated by way of an oral presentation (graduation colloquium) by the candidate and an oral defense with an MSc examination committee. The committee may also include external examiners from research institutes or industrial partners.

If a student undertakes the graduation project at a company, a company-graduation agreement should be signed before the start of the project. Please use the template of the TUD-3mE graduation agreement provided here:

<https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/msc-forms/>

At the start of the literature thesis preceding the graduation project BME students must fill out a BME graduation plan registration form. The template of this form is available on the Brightspace page of the BME Master. There you will also find further instructions on where to send this form.

Program Goals

The BME programme at the TU Delft focuses on three interconnected didactic goals:

To give students a fundamental understanding of the Biomedical Engineering discipline

To train students in the entire process of innovative thinking, design, manufacturing and assessment of biomedical devices

To coach students to perform research in Biomedical Engineering at an academic level

Exit Qualifications

See the course syllabus on the BME Brightspace page.

BME Master 1st Year

Introduction 1

A graduate in Biomedical Engineering is able to (1 to 7):

1. Competent in the scientific discipline Biomedical Engineering
 - 1A. apply a broad and profound knowledge of engineering sciences (mathematics and applied physics) to biomedical problems.
 - 1B. apply general and more advanced but specialized knowledge of anatomy and physiology to general and selected biomedical problems.
 - 1C. apply the knowledge of engineering sciences at an advanced level in at least one Biomedical Engineering specialization.
 - 1D. design, perform, and evaluate experiments.
 - 1E. reflect on standard methods and their presuppositions and to questions these, propose adjustments, and to estimate their implications.
 - 1F. independently spot gaps in own knowledge and to independently revise and extend it through study.
2. Competent in doing research
 - 2A. independently generate new scientific knowledge and new insights within the field of Biomedical Engineering.
 - 2B. assess research, including scientific literature, within Biomedical Engineering on its scientific value.
 - 2C. individually produce and execute a research plan and to choose the appropriate level of abstraction.
 - 2D. deal with the changeability of the research process.
 - 2E. apply a broad knowledge of research ethics.
 - 2F. exercise the best practises in data stewardship and reproducibility.
3. Competent in designing
 - 3A. systematically design complex biomedical systems.
 - 3B. generate innovative contributions to the discipline of Biomedical Engineering.
 - 3C. independently produce and execute a design plan, and to choose the appropriate level of abstraction.
 - 3D. deal with the changeability of the design process.
 - 3E. draw upon other disciplines, especially those from the medical field, in own design.
 - 3F. formulate new research questions on the basis of a design problem.
4. A scientific approach
 - 4A. identify and take in developments in the Biomedical Engineering domain.
 - 4B. critically examine existing theories, models, or interpretations within Biomedical Engineering.
 - 4C. analyse problems and use modelling, simulation, design and integration towards solutions.
 - 4D. document adequately the results of research and design, and to publish these results as to contribute to the development of knowledge in the Biomedical Engineering field and beyond.
 - 4E. reason logically within the field of Biomedical Engineering and beyond, to recognize modes of reasoning, and to recognize fallacies.

See next field for 5 to 7

Introduction 2

(continuation 1 to 4) A graduate in Biomedical Engineering is able to:

5. Basic intellectual skills
 - 5A. analyse and solve technological problems in a systematic way.
 - 5B. identify and acquire lacking expertise.
 - 5C. critically reflect on own knowledge, skills and attitude.
 - 5D. plan and execute research in changing circumstances.
 - 5E. integrate new knowledge in an R&D project, considering ambiguity, incompleteness and limitations.
 - 5F. remain professional with a selflearning ability.
 - 5G. take a standpoint with regard to a scientific argument within the research area.
6. Competent in operating and communicating
 - 6A. work both independently and in multidisciplinary teams.
 - 6B. explain and defend outcomes from the research area to academia and industry, to specialists and peers.
 - 6C. present and report in good English.
7. Considering the temporal and social context
 - 7A. apply a broad knowledge of ethics and statistics in own work.
 - 7B. understand and potentially implement the regulatory procedures required for certification of medical devices relevant to one biomedical engineering specialization.
 - 7C. evaluate and assess the technological, ethical and societal impact of own work.
 - 7D. act responsibly with regard to sustainability, economy and social welfare.
 - 7E. take a note of effective interaction within clinical and pre-clinical settings with clinicians and medical researchers.

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME Obligatory courses (11 ECTS)	
Administration by the Faculty of	3mE

BM41141	Fundamentals of Biomedical Engineering	5
Responsible Instructor	N. Tümer	
Instructor	Ir. A. Sakes	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Course Contents	In this course, there will be: 1) lectures, through which you will become familiar with the field of biomedical engineering (BME) and the stakeholders playing essential roles within, 2) workshops on engineering role play, team dynamics, peer-review, and debate. Through these workshops, you will practice some of the mentioned skills, e.g., being able to work in teams, and having good communication skills, 3) two assignments: (1) self-reflective journey (mostly individual), and (2) challenge-based assignment (group work). These two assignments are connected with each other. In the first assignment, you will explore who you are and what your role would be in relation to your future career in the BME field. In the second assignment, you will have a challenging context. Together with your team, you will define a solution to a challenging problem considering your engineering roles.	
Study Goals	Knowledge, insight, and skills By the end of this course, you should be able to: 1. Define the problems that need to be addressed in the health field, considering crucial societal aspects, and the roles biomedical engineers and stakeholders can play in solving these issues. 2. Apply physical and engineering terms and principles in building a strategy within the multidisciplinary context of biomedical engineering. 3. Outline the steps needed to be taken in order to validate or implement biomedical techniques, devices, and methods in the health system. Transferable skills By the end of this course, you should be able to: 4. Analyse the impact and consequences of your work when intervening in the field of biomedical engineering. 5. Develop evaluative judgment on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback. 6. Reflect on your own interests, role, and personal development in relation to your future professional self in the field of biomedical engineering.	
Education Method	Lectures, workshops and assignments	
Assessment	Exam (40%) and assignments (60%)	
Department	3mE Department Biomechanical Engineering	

BM51007	Professional Development Journey	1
Responsible Instructor	Dr.ir. A.H.A. Stienen	
Instructor	Prof.dr.ir. A.C. Schouten	
Instructor	A.J. Loeve	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	1 2 3 4	
Course Language	English	
Course Contents	Professional development journey ===== In the studentcentered approach of the course Professional development journey (PDJ) students are motivated to explore their own interests, motivators and strength. Students investigate with reflection which moments they felt inspired and what they think is important to make their own choices and create their own learning path. The purpose of the mentorship meetings ===== The mentorship meetings will provide students of BME a space to share ideas and experiences. The sessions will be important to create a safe space for students to share and explore. It is a low stake space where they can talk and think freely. The sessions will be (peer)feedback and reflective oriented meetings to support the reflective assignments for their professional development. In the meetings students help and guide each other to full fill their personal goals. Students can share what they want to get better at and ask for feedback. During the meetings students also listen to their peers and have reflective conversations. Students will also reflect on their own personal goals and development. How they can make the decisions to support their personal growth according to their own goals for the future. Growth mindset (Q3) ===== Most people like the things they are good at. When you get rewarded by good grades or compliments you feel confident and are willing to work harder to get the next reward. In Q3 students explore their growth mindset and reflect on moments of happiness. This to enhance areas of fixed mindset. Reflective conversation (Q4) ===== For this meeting students gather feedback from different people around them (teachers, friends family) and in conversation with a peer about impression students made and how they come across. In the meeting mentors guide a reflective conversation where students evaluate and adjust chosen engineering roles and learning objectives for the decision for an internship/Jip/ or courses. Social context (Q5) ===== In Q5 students are placed in a new social context when joining a Jip or internship. Students make a reflective assignment which focusses on the new social context. Students who are following electives reflect on another moment where they joined a new social context. Graduation plan registration (End Q6/begin Q7) ===== Students prepare their registration and write about the challenges expect for their graduation trajectory and which skills they want to develop? They make a plan for their graduation, check what is necessary (administrative, counselling) and make a SWOT analysis for their graduation project. Mid term evaluation (Three month after start graduation project) ===== Students fill out form 4B to reflect on the progress made in their MSc-thesis, and on planning of the remainder of the project. Students discuss this reflection with their supervisor. Social event (March) ===== Students attend an organized social event for the whole BME department. Guest speakers will be invited to inspire students with their insightful moment.	
Study Goals	The student is able to 1. Reflect on decision making and problem solving: Analyze problems and decisions with reflection methods. Evaluate situations to create different options and select the best. 2. Create a growth mindset: Investigate and explore their own strengths to move from a fixed to a growth mindset. 3. Give and receive feedback: With feedback-seeking behavior and goal orientation stimulate professional development of peers and themselves. 4. Handle contextual and cultural differences: Discover (cultural) differences, cope with differences in team work and is able to choose another perspective.	
Education Method	Each quarter, interactive on-campus mentor sessions are organized. On Brightspace, the topics and assignments will be provided. During the sessions, students can discuss and work on their exercises and ask their peers and mentors for help.	
Assessment	The main assessment of this course is a portfolio assessment which contains six reflective assignments. Graded with Pass/Fail If an assessment is judged as insufficient, the student can write an extended essay on the topic to achieve a passing grade.	
Department	3mE Department Biomechanical Engineering	

WM1402TU	Ethics of Healthcare Technologies	5
Module Manager	Dr. S.M. Copeland	
Co-responsible for assignments	Prof.dr. S. Roeser	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Introduction Ethics and Healthcare Technologies introduces students to the range of ethical issues that tend to arise during the development, design and application of devices used in healthcare or for medical purposes. The class will gain insight into the different aspects of healthcare ethics, such as the distinction between medical research and practice, the ethics of using human or animal research participants, and the ethics of technology, as well as value-sensitive design and responsible innovation approaches within the context of healthcare technologies. The course will provide students with skills in critical reflection, sensitivity to ethical concerns, and the ability to take up considerations of value to society as well as governance in the development and applications of technology. Assignments and cases examined in class will address contemporary developments in these technologies, the reasons behind new regulations that pertain, as well as giving a historical perspective on the ethical principles that underlie current approaches. Thus, the course should prepare students for common problems they will face in their current studies and prepare them for addressing issues that will arise in their future careers. Topics include: Introduction to ethics in medicine and healthcare: The research-practice and normative-descriptive distinctions, Bioethics, and the ethics of clinical research (historical and present), Medical devices and some relevant regulations & codes. Applying ethical frameworks e.g., utilitarianism, principlism, care ethics; Scientific integrity and the social value of research. Introduction to ethics of technology: Technology as a socio-technical system, Innovation in healthcare, Value-sensitive design, responsible innovation. Specific issues in developing healthcare technologies, such as: wearable and other health tracking devices; transhumanism and augmentation; human and animal research ethics; genetic modification; 3D tissue/organ printing; datamining, imagery and related privacy issues; the role of AI, machine learning and robotics.	
Study Goals	Regarding healthcare ethics & ethics of innovation in healthcare, students can: identify and describe various canonical forms of debate and ethical approaches within healthcare and biomedical ethics, compare diverging viewpoints in the field and assess the strength of an argument made in the field, evaluate different ethical issues as they arise at different points along the development and application trajectory. Regarding ethics and technology, students can: identify and describe normative aspects of technology; analyse and evaluate the ethical and social aspects of a case in contemporary healthcare technology development and application, and make recommendations for value sensitive design. Regarding skills, students will have: developed skills in finding and interpreting relevant policy, regulations and codes of ethics for the field; formulated, debated and defended a substantiated position on the ethical issues at stake in current applications of technology in health care.	
Education Method	Contact hours: Plenary sessions that include blended learning activities, 2x Tutorials, Online discussion forums. Study materials: Reader (online) - becomes available during the course; Lecture slides, available on Brightspace after the lecture; Online materials to prepare for in-class activities, including self-directed research.	
Assessment	Grade calculation WM1402 (5ECs): Mandatory attendance in tutorials and short formative assignments, Exam (multiple choice and/or short open questions), Final paper or project (group).	
Remarks	For ethics teaching at TU Delft check https://www.tudelft.nl/ethics/ethics/teaching-activities/ethics-teaching/	
Elective	Yes	
Category	MSc level	

Year	2023/2024
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BME Obligatory Electives: select 1 out of 2, depending on your background (4 ECTS)

BM41055	Anatomy and Physiology	4
Responsible Instructor	Dr.ir. F.J.H. Gijzen	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	In this course, we will focus on the following topics: 1. Introduction to human physiology (human body, homeostasis, mass transport) 2. Cell physiology (plasma membrane, cytoplasm, nucleus, cell growth and reproduction) 3. Musculoskeletal system (bones, joints, muscles) 4. Nervous system (nerve cells, neurophysiology, central nerve system, peripheral nervous system and reflex activity, autonomic nervous system) 5. Endocrine system (hormones and endocrine organs) 6. Cardiovascular system: blood (blood cells, blood flow, blood pressure, vascular compliance, clotting, Newtonian flow). 7. Cardiovascular system: heart (anatomy, coronary circulation, cardiac muscle fibres, cardiac output). 8. Respiratory system (respiration, transport and exchange of oxygen and carbon dioxide, control mechanisms of respiration)	
Study Goals	The student is able to describe the anatomy and the function of several physiological systems. The student must be able to: - Describe the anatomy and the function of the musculoskeletal, nervous, endocrine, cardiovascular, immune, respiratory, digestive, urinary and reproductive systems - Describe the control mechanisms involved in the different physiological systems. For detailed description of learning goals, see first slides of the lectures on Brightspace	
Education Method	1 time per week 2 lectures of 45 minutes Self study before every lesson; cases and examples during (interactive) lectures Course material: Elaine N. Marieb and Katja Hoehn Human Anatomy & Physiology, Global Edition, 10th and 11th edition, Pearson	
Assessment	You will take two digital exams for this course, in Q1 and Q2. These will cover the material of the respective quarters. Both exams account for 45% of your grade. The (digital) resit is in the Q3 exam period and will cover all of the material; this will replace the grade of both exams. In addition, you will take weekly online quizzes which account for 10% of your final grade.	
Department	3mE Department Biomechanical Engineering	

ME41096	Bio Inspired Design	5
Responsible Instructor	Dr.ir. P. Breedveld	
Instructor	B. van Vliet	
Contact Hours / Week x/x/x/x	4.4.0.0	
Education Period	1 2	
Start Education	1	
Course Language	English	
Expected prior knowledge	Mechanics and basic design experience in hand sketching and CAD software, like SolidWorks.	
Course Contents	The course Bio-Inspired Design gives an overview of non-conventional mechanical approaches in nature and shows how this knowledge can lead to more creativity in mechanical design and to better (simpler, smaller, more robust) solutions than with conventional technology. The course discusses a large number of biological organisms with smart constructions and unusual internal mechanisms and presents a number of technical examples and designs of bio-inspired (surgical) instruments, devices, robots, vehicles and machines. Examples of topics: Life and its creation, engineering in biology, strength at low weight, stiffness with soft structures, robustness and redundancy, storing energy in compliant springs, energetically efficient muscle configurations, clamping with hands, claws, suction, glue, dry - and wet adhesion, biological vibration systems, biological walking, swimming, crawling and flying, locomotion of micro- and single-celled-organisms, biological sensors, soft robots, walking robots, flying robots, snake-like surgical instruments and devices. To teach students how to create smart and truly innovative designs, students are trained with the ACCREx creative design method that was developed at the 3mE department Biomechanical Engineering. Alternating intuitive brainstorming with logical, scientific abstracting and categorizing, ACCREx structures and guides a design process towards fundamentally new design solutions. Structure of the course: 1. Engineering & biology - Life and its creation - Finding biological information - Creating creativity 2. Bioconstruction (how are creatures constructed) - Mechanical stiffness & motion - Hydrostatic stiffness & motion 3. Biopropulsion (how do creatures move) - Flying at macro- & microscales - Floating, rolling & adhesive devices - Sliding, crawling & snake-like devices - Walking, compliant devices & compliant actuation 3. Bioclamping (how do creatures grasp) - Friction & biological grasping - Biological hands & prosthetic hands - Biological adhesion	
Study Goals	The student must be able to: 1. describe methods for creative design 2. identify mechanical working principles and phenomena of biological creatures - explain their construction, motion, and/or processing mechanisms - formalize the essence of these mechanisms in models - derive non-conventional design principles from these models 3. implement these design principles in innovative mechanical devices - summarize the transition process from the biological to the mechanical domain - present their design in drawings or preferably in working models	
Education Method	Interactive lectures Interludia with driven scientists in a talkshow-like setting Student presentations about the design process Reflective Q&A sessions	
Assessment	Students are subdivided into multidisciplinary student groups. Each student group receives a design assignment for which a novel, bio-inspired solution has to be found. During the design process, the student groups are coached by the teachers and have to give a number of presentations about their progress to the other student groups. The examination consists of three parts: 1. Each student group hands in a scientific paper covering a problem analysis, the biological background, the design process, the final solution and an evaluation. 2. Each student group gives a final presentation to the teachers and the other student groups. 3. Each student group fabricates a demonstration model that shows the essence of the final solution. The final mark of each individual student is based on a weighted average between these group results and his/her functioning within the student group.	
Department	3mE Department Biomechanical Engineering	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME Elective courses for all tracks
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BM41070	Medical Device Prototyping	6
Responsible Instructor	Dr.ir. T. Horeman	
Instructor	Dr. B.J. van Straten	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	Previous experience in design is a preferred. courses that can help you to prepare are: -BioInspired technology ME41095 -Introduction to Biomedical Engineering BME-MI-087 -Fundamentals of biomechanical Engineering BM41140	
Course Contents	In the course Medical Device Design students develop and produce a sustainable solution to a problem in the medical field in collaboration with their supervisor, clinicians, technical support and sometimes even manufacturing companies. The course offers students to work on a design assignment that encompasses the complete design cycle from problem analysis, development to actual production of the prototype with the support of professional instrument makers. Students will select a design assignment from a number of options proposed by healthcare workers/clinicians (e.g. surgeons, rehabilitation doctors) or companies. The students work in groups of 3 to 4 students under close supervision of an instructor on the assignment, to end up with a working prototype. The course is finalised with a public presentation for clinicians and companies and a report. In bi-weekly meetings of the instructors with all participating groups students are expected to present and discuss the progress in their project in an informal setting.	
Study Goals	The student must be able to: 1. Employ a design task with a multidisciplinary team to solve a real technical problem in a medical environment: translate the clinical problem as presented in the assignment into a practical, technical solution, i.e. do a problem analysis, specify design requirements, come up with a conceptual design, create a cardboard model; obtain feedback on the clinical feasibility of the concept from the medical assignor to further detail the design. bring "green" elements in their design related to reuse, repair, refurbish and recycle to minimise the impact on the environment during production, use and disposal (live cycle) 2. Realize the fabrication of a prototype medical device in collaboration with instrument makers and production companies: create detailed drawings, CAD model; formulate a production plan. 3. Evaluate the performance of the new prototype: test the technical functionality of the device and/or the clinical applicability of the device in a medical setting; reflect on previously made design choices based on the performance of the prototype. 4. Present the design to a multidisciplinary audience of technicians, clinicians and (medical) companies.	
Education Method	Design project	
Assessment	Prototype, report and final presentation	
Remarks	-Students are expected to write a motivation letter before the selection process -Summative feedback is given by a team of experts to guide you towards your end goal.	
Department	3mE Department Biomechanical Engineering	

BM41110	Clinical Patient Safety	2
Responsible Instructor	Prof.dr. M. van der Elst	
Instructor	Prof.dr. J.J. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	During this course, multiple clinical case studies related to incidents in healthcare and related to technology will be evaluated, all in relation to clinical patient safety to provide an optimal clinical healthcare process. Different health technology hazards will be reviewed to discuss possible solutions. These topics cover the effectiveness and efficiency of the healthcare processes, but also the use of technology, its accessibility, and the patient focus in healthcare. At the end, students need to be able to evaluate two out of four specific case studies and provide their findings in a report.	
Study Goals	After successfully finishing this course, the student should be able to: - understand the implication of using technology in healthcare; - analyse a case study and address the core problem; - evaluate the ethical and social impact of their work; - understand regulatory aspects of using technology in this specific setting; - solve a multidisciplinary problem; - effectively interact with healthcare professionals and researchers (within a clinical setting); - present their findings in a concise report.	
Education Method	(Guest)lectures by (medical) specialists from (university) medical centres and the TU Delft, as well as an additional Q&A session.	
Assessment	The students will study cases in groups of four students, preferably from different faculties, and write a report on two of the in total four cases.	
Department	3mE Department Biomechanical Engineering	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-NMB Track NEUROMUSCULOSKELETAL BIOMECHANICS (NMB)	
Introduction 1	We are living in a very demanding society in which people need increasing help to sustain their mobility and vitality throughout their lifetime. This track is focused on understanding the biomechanics of the neuromusculoskeletal system with the aim to improve the vitality of healthy people, enhance the performance of sports professionals and to treat the patients suffering from neuromusculoskeletal disorders and diseases. Therefore, the track is focused on relevant tissue biomechanics, biophysical models of muscles and joints, human motion control, interactions between the biological systems (including brain) and (electro)mechanical devices (e.g., neurorehabilitation robots, haptic interfaces, external prostheses, orthoses) as well as an assessment of the effects of sport equipment on the resultant performance. Students following this track will acquire the fundaments knowledge and research skills for solving increasingly complex and important challenges associated with human movement and control. Courses and research assignments focus on human movement analysis, neuromusculoskeletal modelling, biomechatronics, identification of the (reflexive) control loops in the neuromusculoskeletal system, human-machine interaction, biomaterials, mechanism design, control theory and 3D printing.
Exit Qualifications	FINAL TRACK QUALIFICATIONS NEUROMUSCULOSKELETAL BIOMECHANICS A graduate in Biomedical Engineering Neuromusculoskeletal Biomechanics is able to: 1. Competent in the scientific discipline Biomedical Engineering Neuromusculoskeletal Biomechanics 1a. apply specialized knowledge of tissue biomechanics, biophysical models of muscles and joints, human motion control, interactions between the biological systems (including brain) and (electro)mechanical devices. 1b. apply an assessment of the effects of sports equipment on the resultant performance. 1c. apply medical technologies for the diagnostic and therapeutic purposes of clinical practice on the basis of an integration of knowledge, skills, competence planning and reflection. 1d. apply engineering techniques of biomechatronics and biomechanics to the neuromusculoskeletal body. 2. Competent in doing research 2a. apply research skills for solving complex challenges associated with human movement and control. 2b. formulate research questions and generate innovative solutions in the field of neuromusculoskeletal biomechanics. 3. Competent in designing 3a. develop innovative rehabilitation and treatment procedures and devices based on biomechatronic/biomechanical principles. 3b. develop innovative sports technology or equipment based on biomechatronic/biomechanical principles. 4. A scientific approach 4a. design and perform experiments involving human subjects using sound scientific principles. 4b. critically analyse a complex neuromusculoskeletal problem and translate it into a biomechatronic/biomechanical solution. 5. Basic intellectual skills See general BME final qualifications 6. Competent in operating and communicating 6a. create/maintain a respectful subject-experimentalist relationship within the bounds of human subject-experimentalist interaction. 7. Considering the temporal and social context 7a. optimize the interaction between biological and mechanical systems and generate new/innovative technical solutions.

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-NMB Obligatory (25 ECTS)

Introduction 1

Please note that students that have a background in Biology, Biomedical Engineering/Technology, Biomedical Sciences, Biopharmacy, Clinical Technology, Human Movements Sciences, Medical Natural Sciences, Medicine, or Veterinary Medicine are not allowed to follow the course Anatomy & Physiology [BM41055]. They have to choose another course instead.

BM41040	Neuromechanics & Motor Control	5
Responsible Instructor	Dr.ir. W. Mugge	
Instructor	Prof.dr. F.C.T. van der Helm	
Instructor	Prof.dr.ir. A.C. Schouten	
Instructor	Prof.dr. H.E.J. Veeger	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Relevant TU Delft BSc Courses: -Wiskunde 2 Lineaire Algebra WBMT1051 -Wiskunde 3 Differentiaal Vergelijkingen WBMT2048 T2 -Wiskunde 4 Kansrekening & Statistiek and Numerieke Wiskunde WBMT2049 T1&T2 -Rigid Body Dynamics WB2630 -Signaalanalyse WB2235 -Systeem- en Regeltechniek WB3240	
Course Contents	Humans have an extraordinary flexibility in generating movements. With little attention most of us can perform complex tasks such as walking or riding a bicycle. The central nervous system integrates available sensory information and select the proper motor command for the muscles. The skeletal system enables the transfer of forces required for body movement while the muscles produce the forces to accelerate it. Stability is maintained through both passive (intrinsic muscle properties) and active (reflexes) contributions to the visco-elasticity of the joints. In this course we investigate how humans generate movements, stabilize posture, and integrate sensory feedback into control actions, while acknowledging the kinematics and the mechanical properties of the human motor system. This course addresses fundamental aspects and clinical applications. During the course the role of subsystems (e.g. muscles, sensory systems) and their interactions as well as uncertainty in sensory signals are discussed with a focus on reaching movements and posture maintenance. During the course it will be shown how this knowledge can help in understanding impaired motor function after neurological disorders, like Stroke and Parkinsons disease.	
Study Goals	After this course the student must be able to: - Describe the motion of bony segments of the human skeleton system with respect to each other using kinematic analysis methods. - Calculate muscle forces from known joint motion using different optimization techniques (forward dynamics and inverse optimization). - Describe the physiology of the human motor and sensory systems as well as the central nervous system with respect to motor control. - Build and use neuromusculoskeletal models to simulate movements using motion equations. - Explain the neuromuscular system from a control engineering perspective, including the interaction between intrinsic and reflexive feedback. - Describe how the central nervous system deals with uncertainty, integrates sensory feedback and adapts to new situations. - Is able to deduce how impairments of different processes involved in motor control due to neuromuscular disorders affect motor control and adaptation. - Is able to critically judge the value of theories about postural and movement control posed in scientific literature.	
Education Method	Lectures and practicals (4 hours per week) and homework assignments	
Assessment	Written/Digital exam, computer tests (digital midterm) and assignments	
Tags	Lab Research Matlab Mechanics Mechatronics Modelling Optimalisation Signals and Systems Technology	
Department	3mE Department Biomechanical Engineering	
Contact	Preferably by email to the Responsible Instructor	

BM41090	Computational Mechanics of Tissues and Cells	6
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Instructor	N. Tümer	
Instructor	M.J. Mirzaali Mazandarani	
Contact Hours / Week x/x/x/x	0/0/3/3	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	This course is required for students following the Neuromusculoskeletal Biomechanics track of the Biomedical Engineering master programme. Other students are more than welcome to enroll.	
Course Contents	The course consists of two parts: lectures and hands-on workshops. The lectures are designed to give students in-depth knowledge of the mechanical behaviour of cells and tissues and how their behaviour can be explained in terms of mechanical theories. The hands-on workshops give students the opportunity to apply what they have learned in the lectures. A commercial Finite Element (FEM) modelling package is used to make FEM modelling as practical as possible. The emphasis is therefore shifted from the technical details of the FEM modelling towards understanding the modelling procedures and applying them for analysis of certain problems. The problems are chosen to be as close to the real-world problems as possible. In addition to finite element modelling, the principles of medical image processing for finite element modelling are discussed. The course is divided into two almost equal parts. The first part covers patient-specific finite element modelling of tissues and implants. The second part is focused on the mechanical behaviour of cells and proteins (such as collagen). The following topics will be covered in the lecturing (tentative): Lecture 1: Introduction to the course Lecture 2: Geometry of tissues and implants for finite element modelling (including statistical shape and statistical shape and intensity models) Lecture 3-5: Material modelling and material properties (including isotropic and anisotropic linear elasticity, hyperelasticity, viscoelasticity, and poroelasticity) Lecture 6: Loads and boundary conditions (including musculoskeletal models) Lecture 7: Modelling of tissue growth and adaptation Lecture 8: Introduction to cell mechanics, continuum models of the cell Lecture 9: Tensegrity models of the cell Lecture 10: Polymer-based models of the cell Lecture 11-12: Protein mechanics The workshops start from very simple modelling exercises that introduce the FEM package and show how it can be used for simple modelling tasks. The level of difficulty will gradually increase up to the point that students can do FEM modelling of tissues, cells, proteins, and tissue-implant systems.	
Study Goals	Students who have successfully completed this course must be able to: 1.Explain continuum theories that are used for modelling of biological tissues (i.e.anisotropic elasticity, viscoelasticity, and poroelasticity) 2.Translate physiological/pathological conditions of cells and tissues to the language of mechanics 3.Comfortably use a certain commercial FEM package (possess general FEM modelling skills) 4.Build FEM models of tissues and cells 5.Identify different modelling approaches 6.Make appropriate choices concerning material models, boundary conditions, loadings, etc. 7.Critically assess the results of their FEM models	
Education Method	(Interactive) lectures and hands-on workshops	
Computer Use	Computer will be used for the hands-on workshops	
Literature and Study Materials	Recommended texts for the course: 1.Cowin, Stephen C., Doty, Stephen B., 2007, Tissue Mechanics, Springer. 2.Cowin, Stephen C. (Editor), 2001, Bone Mechanics Handbook 2nd Ed., CRC Press. 3.Bartel, Donald L., Davy, Dwight T., Keaveny, Tony M., 2006, Orthopaedic Biomechanics: Mechanics and Design in Musculoskeletal Systems, Prentice Hall. 4.Mow, van C., Huiskes, Rik, 2004, Basic Orthopaedic Biomechanics & Mechanobiology 3rd Ed., Lippincott Williams & Wilkins. 5.Jacobs, Christopher R. et al., 2012, Introduction to Cell Mechanics and Mechanobiology, Garland Science.	
Prerequisites	Students should have working knowledge of the mechanics of materials	
Assessment	Final exam (for the lecturing part) + submission of lecture (individual) and workshop (group) assignments: The grading policy for this course includes: Workshop assignments (total: 60%, first two assignments: 15% each, last workshop assignment: 30%). Lecture assignments (total: 15%, each assignment: 5%) Written exam (25%) 50% of the grade of the written exam is needed to pass the course. At least 2 out of 3 lecture assignments have to be submitted within the deadline provided in the course to pass the course The three workshop assignments are team work activities, and the final report has to be approved by all group members. Once the group members are defined, they cannot be changed. In case of any problems with your group members, we preserve the right to ask for an oral examination. If students fail the re-sit exam, in order to pass the exam in the next academic year, they are asked to not only retake the exam but also submit again two lecture assignments. Any delays with the submission of the lecture assignments will result in down grading of your grade. No delays in the submission of workshop assignment is accepted.	
Department	3mE Department Biomechanical Engineering	

BM41131	Tissue Biomechanics	3
Responsible Instructor	A.C. Akyildiz	
Instructor	Dr.ir. M.P. Peirlinck	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course involves lectures and interactive group discussions. In the lectures, the students will be introduced to the biomechanics of various tissue types in the human body, including bone, cartilage, tendon, ligament, muscle, and vessel tissues. The students will be guided through the topics of these tissue types focusing on structure-function relationship, material behavior (experimental examples), combined with modeling approaches and clinical application. The fundamental knowledge on continuum mechanics, required for a good understanding in tissue biomechanics, will be provided as well. Additionally, experts will be invited to talk about their current scientific research on tissue biomechanics, representing the rapidly developing field of tissue biomechanics. During the group discussions, the students will be expected to read and take a deeper dive into selected tissue-specific research articles and discuss the topics background, approach, methods, results, and limitations with their peers. Active participation in these group discussions is obligatory in order to take the final examination.	
Study Goals	After successful completion of this course, the student should be able to: 1. Explain the biomechanical constituents, hierarchical structure and mechanical function of biological tissues, and how they relate to each other, 2. Discuss the basic concepts of continuum mechanics that are relevant for tissue biomechanics, 3. Examine the mechanical behavior of biological tissues using continuum mechanics framework, 4. Evaluate recent scientific research on tissue biomechanical modeling and experimentation, and its applicability in the clinic.	
Education Method	Lectures, group discussions	
Literature and Study Materials	Book / online material recommended to complement lectures: An introduction to biomechanics: solids and fluids, analysis and design (Jay D Humphrey & Sherry L O'Rourke) Nonlinear solid mechanics: a continuum approach for engineering (Gerhard Holzapfel) Continuummechanics.org	
Assessment	Group assignments Written exams	
Department	3mE Department Biomechanical Engineering	

ME41065	System Identification and Parameter Estimation	7
Responsible Instructor	M.L. van de Ruit	
Instructor	Prof.dr.ir. A.C. Schouten	
Instructor	Dr.ir. W. Mugge	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic control engineering and experience with Matlab	
Course Contents	System identification deals with the estimation of physical properties of (dynamic) systems from measured data (inputs and outputs), which is essential for the understanding of system behavior. System identification is particularly useful for biological systems where physical entities are generally not known a priori. The course provides mathematical a background of system identification in both time domain and frequency domain with a strong focus on practical application. Major topics to be studied are: spectral densities, nonparametric black-box models, discrete time domain models (parametric grey-box), closed loop systems (utilizing feedback), translation of transfer functions into physical parameters, optimization routines and design of perturbation signals. During the course examples from both technical and physiological systems (i.e. estimate the behavior of a driver or the dynamics of a human joint) will be discussed.	
Study Goals	At the end of the course the student is able to. LO1: Signals a: explain the importance of input signal design and signal acquisition properties for the identification of system dynamics. b: select optimal input signals to identify unknown system dynamics and test them using MATLAB and Simulink. LO2: Systems a: illustrate how system dynamics can be accurately estimated using non-parametric (non)linear time-invariant models in time- and frequency domain. b: estimate nonparametric models of unknown systems from measured input and output signals using MATLAB and Simulink. LO3: Models a: explain what model structure is best fits a non-parametric estimate and which corresponding optimization algorithm to use to find the model parameters. b: parameterize physical models from nonparametric estimates employing optimization techniques using MATLAB and Simulink.	
Education Method	Self study, lectures (2x2 hours per week) including (home) work assignments	
Prerequisites	recommended prior knowledge - Probability theory and statistics (e.g. WBMT2049-T1) - Signal analysis (e.g. WB2235) - Systems and control engineering (e.g. WB3240)	
Assessment	Final closed book exam (60%), digital midterm (40%) tests and assignments (pass/fail)	
Tags	Analysis Mathematics Matlab Mechanics Modelling Numeric Methods Programming Signals Signals and Systems Small groups Technology	
Department	3mE Department Biomechanical Engineering	
Contact	By email SIPE-3ME@tudelft.nl	

ME41085	Biomechatronics	4
Responsible Instructor	Dr.ir. G. Smit	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	Biomechatronics is a contraction of biomechanics and mechatronics. In this course the central focus is on function and coordination of the human motion apparatus, and on the design of assistive devices for the support of the function of the motion apparatus. Examples are assistive devices like an orthosis, a prosthesis or Functional Electrical Stimulation of muscles. The goal is to provide some function to patients with functional deficiencies.	
Study Goals	After successfully completing this course, you are be able to: 1. make a Problem Analysis of a given assignment in the field of the human motion apparatus and its interaction with an assistive device a. identify the underlying cause [pathology] of the problem as presented in the assignment b. describe and explain the possible treatment options for the pathology of 1a c. translate the result of 1a into technical Design Objectives without reference to any solutions d. derive qualitative and quantitative Design Specifications from the Design Objectives, and categorize and prioritize these 2. optimize the assistive device application given in the assignment in energetic and control aspects a. select and apply appropriate Design Methodology and Design Methods b. generate a variety [typically at least three] of Conceptual Designs c. judiciously select the most appropriate Conceptual Design d. demonstrate the plausibility or feasibility of the Conceptual Design, with special emphasis to the patient benefits	
Education Method	Lectures (2 hours per week)	
Literature and Study Materials	Course material: A reader is available through brightspace References from literature: D.B. Popovic and T. Sinkjaer Control of Movement for the Physically Disabled Springer (2000) ISBN-13: 978-1852332792 D.H. Plettenburg Upper Extremity Prosthetics. Current status & evaluation VSSD (2006) ISBN-13: 978-9071301759	
Prerequisites	Assumed prior knowledge when entering the course, a basic understanding of: Statics Dynamics Control Theory Engineering Design	
Assessment	Assignment: Report made in a group (~2 persons)	
Remarks	Old course code: WB2432	
Department	3mE Department Biomechanical Engineering	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-NMB Obligatory electives: Select 3 courses from the following list

BM41035	Biomaterials	4
Responsible Instructor	Dr.ir. I. Apachitei	
Instructor	Dr.ir. E.L. Fratila-Apachitei	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In improving patient care, especially in relation with tissue loss or dysfunction, the use of proper biomaterials plays a key role. The shift from tissue removal to tissue replacement and at present, tissue regeneration is driven by the (i) evolution of biomaterials from bioinert to bioactive and bioresorbable and (ii) the increasingly complex biomedical problems of an aging and more active population. As an example, the existing total joint replacements have a 75 - 85 % survivability at 15 years. However, since they were first introduced in the 1960s, average life expectancy increased by at least ten years impacting on the quality of the bone and implants survivability that deteriorate with age. As a consequence, improved implant survivability by 10 - 20 years and alternative treatment methods that delay or eliminate the use of prostheses are badly needed. The Biomaterials course covers in an interdisciplinary approach (biomaterial engineering and biology) the background information on type, composition, processing and properties of biomaterials along with their evolution. Metallic, polymeric and ceramic materials are included and their properties are described and compared with those of biological materials. The main methodologies for biomaterials characterization are covered as well as the procedures for biomaterials standardization and regulation. Further, the interactions between biomaterials and specific biological environments are explained and used for defining the specific requirements and selection principles for biomaterials. Here, biomaterials and medical devices for orthopedics, cardiovascular system and tissue engineering are covered. Finally, the research directions and approaches for development of novel biomaterials are outlined.	
Study Goals	At the end of the course the students will be able to: 1. Identify biomaterials used in: orthopedics, cardiovascular implants, and scaffolds for tissue engineering 2. Define the properties of biomaterials and their property map (range of property values) 3. Describe structure-property relationships for various biomaterials 4. Describe the methodologies for biomaterials characterization and regulation 5. Compare and contrast biomaterials and biological materials 6. Explain the main interactions between a biomaterial and a biological environment 7. Describe the principles for biomaterials selection and design 8. Describe approaches and processes used for functionalization of biomaterials 9. Identify, analyse and describe a biomaterial problem associated with a medical device 10. Explain the clinical performance of biomaterials and medical devices	
Education Method	Lectures, case studies and workshops	
Literature and Study Materials	Lecture slides and recommended literature (reader)	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

BM41155	3D Printing	4
Responsible Instructor	Dr. J. Zhou	
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Instructor	Ing. M.A. Leeflang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	in Track I (Musculoskeletal Biomechanics)and Track II (Medical Devices and Bioelectronics) of MSc Biomedical Engineering BME) and in the specialization of Bio-Inspired Technology (ME-BMD-BITE)	
Expected prior knowledge	BSc in mechanical engineering, materials engineering, industrial design engineering and aerospace engineering with a basic knowledge of materials (e.g. WB2330)	
Parts	- Overview of AM technologies, their principles and applications - Materials selection (polymers, ceramics, metals and composites) - Powder-bed fusion processes - Powdered materials and powder fabrication - Powder characteristics in relation to AM - Structural integration during melting and sintering - Extrusion-based systems - Design for functionality and 3D printability Software issues and AM limitations - Other powder-based manufacturing technologies as alternatives to AM - Post-processing of AM products - Mechanical testing of AM products - Case studies AM for engineering and medical applications - Assignment and presentations	
Course Contents	Additive manufacturing (AM), also known as 3D printing the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to conventional subtractive manufacturing technologies, is set to revolutionize manufacturing across a wide range of industries. The course 3D printing - AM technologies is designed to equip students with the fundamental knowledge of AM and basic hands-on experience in AM. The topics include various AM processes, their principles, capabilities and limitations, materials used for AM processes, powdered material fabrication and characterization, CAD designs for AM, alternative powder-based manufacturing technologies, current and future AM applications, as well as the mechanical evaluation of AM parts. The primary focus is placed on powder-based AM for metallic parts. In addition to a series of lectures, project groups, each composed of 3-5 students, will create their own designs of the objects made of polymer PLA, formulate requirements in terms of geometry, dimensions, tolerances, mass, surface finish, functionality and mechanical properties of these objects, redesign them by making use of SolidWorks, print them, test and evaluate 3D printed products against the requirements, and document the projects in the form of reports and presentations. The new design is expected to reflect the distinctive features and freedom that AM has to offer (thin walls, lattice structures, form-free channels, integrated parts/functions, and customization), but within the capacity of the printing machines (bounding box size, minimum feature size or wall thickness). The report and presentation are expected to contain: reason for the selection of the object, weakness of the existing design, new design features, actual testing results and recommendations.	
Study Goals	Students will be familiar with the concept of AM, various AM processes, AM applications, materials for AM, product design for AM, post-processing techniques and evaluation methods. In addition, students will gain basic hands-on experience in product development via AM. Once successfully completing the course, the students will: - Have a fundamental knowledge of 3D printing/AM technologies for a wide range of applications - Be able to explain the benefits, advantages, challenges and limitations of individual AM technologies as well as alternative powder-based manufacturing technologies - Have an insight into selective laser sintering and melting (SLS/SLM) processes regarding process control in relation to the geometry and mechanical properties of the products - Become familiar with the common (metallic) materials for 3D printing and their characterization techniques - Understand printable geometry design rules and the common methods of digital file creation - Become familiar with the common techniques for post-processing - Become familiar with the common methods for the mechanical evaluation of 3D printed materials and functional parts.	
Education Method	Lectures and student projects	
Computer Use	Attending lectures and performing assignment are both compulsory	
Computer Use	- Object design with Software (SolidWorks Student Engineering Kit)	
Course Relations	Biomaterials, Tissue biomechanics and other engineering courses in MSc biomedical engineering, materials science and engineering, mechanical engineering, marine technology, aerospace engineering, design for interaction, integrated design and strategic design	
Literature and Study Materials	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Books	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21)	
Reader	- Lecture materials	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

BM41170	Physiological Flows and Transport	4
Responsible Instructor	Dr. S. Pirola	
Instructor	F. Fontana	
Instructor	Dr.ir. W.J. Hogendoorn	
Instructor	Dr.ir. F.J.H. Gijzen	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The aim of the course is to introduce the basic principles and concepts of fluid flow and mass transport in physiological systems. The course is designed to develop a quantitative understanding of the underlying physical and biological phenomena governing flow and mass transport in the human body. These aims will be achieved through conventional lectures, guest lectures on relevant topics from experts in the field, problem solving classes (involving problem sheets to be solved at home and discussed in class), and a workshop on computational fluid dynamics.	
Study Goals	By the end of this course the students will be able to - Explain key principles of fluid dynamics and mass transport. - Apply these principles to describe the fluid flow and mass transport in physiological systems. - Solve problems involving fluid flow and mass transport in physiological systems. - Discuss basic concepts of computational fluid dynamics.	
Education Method	Lectures, problem sheets, workshops	
Literature and Study Materials	Lecture notes/slides Biofluid Dynamics: principles and selected applications (C. Kleinstreuer) Transport Phenomena in Biological Systems (G. Truskey, F. Yuan, D. Katz)	
Assessment	Written exam	
Tags	Fluid Mechanics Transport phenomena	
Department	3mE Department Biomechanical Engineering	

ME41006	Musculoskeletal Modeling and Simulation	4
Responsible Instructor	A. Seth	
Instructor	Ir. J.D.C. Groenhuis	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Human and animal movement is often smooth, powerful, and graceful. Animal movement is generated through a cascade of nervous, muscular, and skeletal system dynamics. Even the simple tasks of walking and running require the coordination of hundreds of muscles that pull on bones that move our limbs and generate forces on the environment so we can move. When these systems are disrupted through disease and injury, they result in movement disorders that can be debilitating. In this course you will learn to construct and simulate musculoskeletal models that generate movement. You will learn how to apply models to improve measurements of movement and to predict behavior, thus extracting more insights about human (and animal) movement than from observations alone. We will start with the constituent components of bones, joints, muscles, and tendons and their dynamics, as the building blocks of musculoskeletal systems. You will learn inverse techniques to extract unmeasured quantities, like joint kinematics, work, and power from observations and to use forward dynamic simulations to predict behaviors and answer what if? type questions. You will apply these models and techniques to answer a research question of your own formulation	
Study Goals	The goal of this course is to teach you how to use state-of-the-art computational tools to model and simulate the human musculoskeletal system. Learning will be guided by lectures, reading articles, practical labs and a final project. Upon successfully completing this course, you will be able to: 1. Model Develop and analyze mathematical models of the human musculoskeletal system using OpenSim and MATLAB. 2. Simulate Create inverse and forward dynamic simulations of human movement. Perform inverse dynamic simulations and optimizations using experimental data to compute quantities that cannot be easily measured (e.g., muscle forces). Perform forward dynamic simulations to reproduce observed movements and predict new movements. 3. Analyze Use musculoskeletal simulation to answer questions about human movement (e.g., how does a muscular impairment affect muscle coordination?), and validate results against experimental data. 4. Critique Read journal papers in the field; establish the main contributions and limitations of the work. Summarize your work using a thoughtful structure and relate your work to the broader field. You will also review the work of your peers and provide feedback on clarity, correctness and impact. 5. Formulate a research question and write a corresponding proposal.	
Education Method	Flipped Classroom method. Lecture slides, reading materials, and assignment descriptions provided through Brightspace. Class time is used to review reading materials and answer questions on concepts and practical labs. The bulk of the work is performing challenge based labs and a final project of your choice.	
Assessment	60% Three practical computer lab assignments (20% each) 40% Final project (proposal, presentation, abstract, reproducibility, reviews) Pass 6/10 grade total	
Department	3mE Department Biomechanical Engineering	

ME41035	Special Topics in Sports Engineering	3
Responsible Instructor	J.K. Moore	
Contact Hours / Week x/x/x/x	x/x/x/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	*WARNING* This course is an intensive, full time 2 week summer course that will occur during the months of June, July, or August. If in June, it will conflict with your other courses so plan accordingly for the work load. Check the course website for up-to-date information: https://moorepants.github.io/me41035/ Special Topics in Sports Engineering is an inter-university course for Master students in Mechanical Engineering, Movement Sciences, Sport Sciences, and other related MSc programs. The course is organized as a two-week intensive course and comprises lectures, demonstrations, practicals, hands-on research, and a final field test. The course will be taught by staff from the six participating universities. Due to the ongoing coronavirus situation, this year's course will again be online. The course is organized around a basic theme relevant for sports engineering and this year's theme will be "Maximizing cycling performance". During the course students will work out what aspects determine cycling performance, and collect data (through experiments or literature research) that are needed to develop and feed a simulation program for the estimation of the optimal bicyclerider combination and the maximal performance humanly possible. The course's final activity will be a test ride to quantify the differences between actual performance and predicted performance. In this course, students will have to answer the question: Given a particular bike, what will be your own predicted time over a given distance and track and how well does this match reality? The prediction should be based on a power-based simulation model of cycling and the relevant bicycle and rider dependent parameters, which have to be collected experimentally. Answering this question will require insight in relevant parameters, but also collecting these parameters, for each individual student with his or her individual bike. The bike in question can be chosen freely and might therefore be a top-end racing bike as well as your grandmother's shopping bike.	
Study Goals	After following this course, students should understand the complexity of maximizing sports performance and the importance of the inclusion of materialathlete interaction. More specifically, students should be: - Familiar with the Power Equation concept and be able to apply this to cycling; have knowledge of methodological aspects of sports research, in particular error propagation, manmachine interaction (closed loop complexity), measurement techniques, internal and external validity. - have insight in the organizational and psychological complexities of sports innovation. - Able to measure key parameters needed for power equations, related to their own field and have experience in the measurement of key parameters in adjacent fields; - able to provide a cycling performance simulation program with the parameters necessary to evaluate performance on a realistic level; - able to collect and present to fellow group members, data on parameters for such a simulation program. - present research findings through an individual portfolio, and a group presentation/poster/brief oral.	
Education Method	lectures, workshops, practicals, presentations, team work	
Assessment	A pass/fail grade will be given based on participation in all practicals, presentations, writing reports, and a peer review.	
Department	3mE Department Biomechanical Engineering	

ME41056	Multibody Dynamics	5
Responsible Instructor	J.K. Moore	
Instructor	W. Wolfslag	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3	
	4	
Start Education	3	
Exam Period	4	
	5	
Course Language	English	
Department	3mE Department Biomechanical Engineering	

RO47006	Human Robot Interaction	5
Responsible Instructor	Prof.dr.ir. J.C.F. de Winter	
Instructor	Dr.ir. D.A. Abbink	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course on Human-Robot Interaction aims to equip students with the knowledge necessary to design robots that can function safely, comfortably, and effectively in environments inhabited by humans. The course offers both theoretical and empirical insights into the physical and cognitive aspects of human-robot interaction. Students will learn to model the fundamental principles of this interaction, critically examine existing theories and models, and assess various design options by reflecting on and experiencing experimental research.	
Course Contents Continuation	The course will cover key principles of human information-processing, human error, the interaction between humans and automation, human-machine interfaces, and operator state assessment (which includes factors like visual attention, workload, and situation awareness). The lectures will delve into the performance of human operators in roles ranging from manual control (direct control) and supervisory control of automated systems, to shared control of human-robot systems. Additionally, students will learn essential elements of experimental design for human-subject studies. The course finds its main applications in areas such as intelligent vehicles and industrial robotics. To supplement the core curriculum, guest lectures will address specialized topics like haptic interfaces, motion comfort in automated vehicles, eye movements in supervisory control, and collaborative human-robot interactions.	
Study Goals	Upon completing this course, students will be able to: - Explain the mechanisms of human information processing, which includes sensation, perception, decision-making, and action execution. - Explain the key theories and models related to human error, learning, and individual differences. - Describe and model direct human interactions with robots and other machines. - Articulate how humans supervise and interact with robots at higher levels of automation. - Critically examine concepts such as trust, overreliance, and behavioural adaptation in human-robot interaction. - Describe and reflect on the design of interfaces that facilitate effective human-robot cooperation. This includes classical interface design principles, but also more modern haptic interfaces as well as linguistic interfaces such as ChatGPT. - Adapt robotic functionality to human needs, either by teaching the robot or through operator state assessment. - Process experimental data and bio-signals to infer human visual attention, workload, and situation awareness. - Design and critically evaluate experimental studies involving human subjects for the assessment of robotic systems. - Provide constructive feedback to peers.	
Education Method	The course is taught through lectures, totalling four hours per week. Some of the lecture hours will be practical or interactive in nature. In addition to the regular lectures, there will also be guest lectures. Participation in sessions related to practical exercises is obligatory. Students will also participate in an experiment.	
Literature and Study Materials	The course materials will be made available on Brightspace. This includes: (1) the slides, (2) notes accompanying the lectures, and (3) scientific articles and book chapters. Example literature: - Parasuraman, R. (1997). Humans and automation: Use, misuse, disuse, abuse. <i>Human Factors</i>, 39, 230253. https://doi.org/10.1518/001872097778543886 - De Winter, J. C. F., Petermeijer, S. M., & Abbink, D. A. (in press). Shared control versus traded control in driving: A debate around automation pitfalls. <i>Ergonomics</i>. https://doi.org/10.1080/00140139.2022.2153175 - De Winter, J. C. F., & Dodou, D. (2017). Human subject research for engineers: A practical guide. <i>SpringerBriefs in Applied Sciences and Technology</i>. https://doi.org/10.1007/978-3-319-56964-2 - Mulder, M., Pool, D. M., Abbink, D. A., Boer, E. R., Zaal, P. M. T., Drop, F. M., Van Paassen, M. M. (2017). Manual control cybernetics: State-of-the-art and current trends. <i>IEEE Transactions on Human-Machine Systems</i>, 48, 468485. https://doi.org/10.1109/THMS.2017.2761342	
Assessment	Exam (45% of the final grade): This will be an on-campus, open-book digital examination. Additionally, the course incorporates two homework assignments: Assignment 1 is a group task related to manual control, while Assignment 2 is an individual task related to supervisory control of robotic systems and human-state assessment. Additionally, students will participate in a human-subject experiment.	
Department	3mE Department Cognitive Robotics	

SC42001	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation. Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design. Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. Due to COVID-19 the exam may be an online exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Department	3mE Department Delft Center for Systems and Control	

SC42015	Control Theory	6
Responsible Instructor	T. Keviczky	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students following this course are expected to have a strong background in the following topics: - Linear algebra (basic concepts up to generalized eigenvectors and Jordan canonical form) - Calculus (smoothness, Jacobian, gradient) - Differential equations (existence, uniqueness, Laplace transform) - Classical control (transfer functions, block diagram algebra, Bode/Nyquist, etc.)	
Course Contents	- State-space description of multivariable linear dynamic systems, interconnections, block diagrams - Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation - Dynamic response, relation to modes, the matrix exponential and the variation-of-constants formula - Realization of transfer matrix models by state space descriptions, coordinate changes, normal forms - Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback - LQ regulator, robustness properties, algebraic Riccati equations - Observability, detectability, unobservable modes, state-estimation observer design - Output feedback synthesis (one- and two-degrees of freedom) and separation principle - Disturbance and reference signal modeling, the internal model principle	
Study Goals	The student is able to apply the developed tools both to theoretical questions and to simulation-based controller design projects. More specifically, the student must be able to: - Translate differential equation models into state-space and transfer matrix descriptions - Linearize a system, determine equilibrium points and analyze local stability - Describe the effect of pole locations to the dynamic system response in time- and frequency-domain - Verify controllability, stabilizability, observability, detectability, minimality of realizations - Sketch the relevance of normal forms and their role for controller design and model reduction - Describe the procedure and purpose of pole-placement by state-feedback and apply it - Apply LQ optimal state-feedback control and analyze the controlled system - Reproduce how to solve Riccati equations and describe the solution properties - Explain the relevance of state estimation and build converging observers - Apply the separation principle for systematic 1dof and 2dof output-feedback controller design - Build disturbance and reference models and apply the internal model principle	
Education Method	Lectures and Exercise Sessions	
Computer Use	The exercises will be partially based on Matlab in order to train the use of modern computational tools for model-based control system design.	
Course Relations	This course is intended primarily for MSc students that are enrolled in the Systems & Control MSc program! For students who are interested in Systems & Control within their own MSc program, the SC42001 Control System Design course is recommended instead. There is a large overlap between the two courses in terms of topics, but SC42001 emphasizes broader concepts of control, it is more application-oriented, and does not discuss the technical proofs and mathematical background so extensively.	
Literature and Study Materials	B. Friedland, Control System Design: An Introduction to State-space Methods. Dover Publications, 2005 K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton and Oxford, 2009 http://www.cds.caltech.edu/~murray/amwiki/index.php?title=Main_Page	
Assessment	Written mid-term examination (15% grade) and written final examination (85% grade) will determine the total final grade. The mid-term result is only valid for the academic year when it was obtained. For the resit examination there will be a written examination (100% grade) for which the mid-term result will not count. If on-campus exams will not be possible, then all exams will be organized online (via Ans).	
Department	3mE Department Delft Center for Systems and Control	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-MD Track MEDICAL DEVICES (MD)

Contact for students	Prof.dr.ir. W. A. Serdijn Email: w.a.serdijn@tudelft.nl
Program Title	MSc Biomedical Engineering
Director of Education	Prof.dr. Amir Zadpoor Faculty of Mechanical, Maritime and Materials Engineering (3mE) Department of Biomechanical Engineering Section Biomaterials and Tissue Biomechanics Email: A.A.Zadpoor@tudelft.nl Room: B34-E3-290
Program Coordinator	Arjan Loeve Faculty of Mechanical, Maritime and Materials Engineering (3mE) Department of Biomechanical Engineering Email: a.j.loeve@tudelft.nl Room: F-0-200
EC Program	120 ECTS, two years Master of Science program
Introduction 1	This track provides an integrated platform to enable development of advanced medical devices including biomaterials, design models and fabrication processes for implantable devices, biosensors, medical instruments, external prostheses, orthoses, as well as diagnosis and disease monitoring systems. From novel multifunctional biomaterials to device design and prototype fabrication, and finally testing of the devices in clinically relevant environments, the students are provided the needed theoretical knowledge and are then challenged to prove their concepts by working in state-of-the-art research laboratories. Multidisciplinary teams of teachers and researchers are involved in this track to integrate the relevant topics and provide a focused program. A wide range of medicals devices is covered. That include: biomaterials and implants able to repair and regenerate diseased tissues, fabricated by integrated technologies involving 3D printing, micro/nanofabrication and surface biofunctionalization, instruments for minimally invasive surgeries, virtual reality trainers and simulators with force/haptic feedback, prostheses, flexible and stretchable electronics and sensor microsystems for medical devices are in the focus of the education and research program of this track. As already confirmed, the students following this track acquire unique multidisciplinary expertise that makes them ready to face the challenges encountered by the biomedical companies active in these fields as well as by the clinicians and research teams in hospitals, academia and research institutes worldwide. Such specialists represent the guarantee for finding the solution of tomorrow related to medical devices for healthcare systems.
Exit Qualifications	FINAL TRACK QUALIFICATIONS MEDICAL DEVICES A graduate in Biomedical Engineering Medical Devices is able to: 1. Competent in the scientific discipline Biomedical Engineering Medical Devices 1a. apply a broad and profound knowledge of design, biomedical materials, experimental methods, manufacturing technologies, biomedical microsystems and biomedical electronics to medical devices. 1b. apply general and more advanced but specialized knowledge of anatomy and physiology to medical devices. 1c. apply the knowledge of engineering sciences at an advanced level in at least one type of medical device. 1d. apply the methodologies for medical device development and approval (regulatory issues). 2. Competent in doing research 2a. independently generate new scientific knowledge regarding the interaction of medical devices with the human body. 3. Competent in designing 3a. translate a complex clinical challenge in an elegant multifunctional technical solution. 3b. can apply and evaluate the various steps involved in the whole cycle of medical device design. 3c. apply medical device design in clinically relevant areas. 3d. generate innovative contributions to the discipline of medical devices. 3e. optimize the interaction between biological, mechanical and electronic embedded systems. 4. A scientific approach See general final qualifications BME 5. Basic intellectual skills See general final qualifications BME 6. Competent in operating and communicating See general final qualifications BME 7. Considering the temporal and social context 7a. apply the latest design strategies for circular medical products and developments.

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-MD Obligatory courses (24 ECTS)

Introduction 1

Please note that students that have a background in Biology, Biomedical Engineering/Technology, Biomedical Sciences, Biopharmacy, Clinical Technology, Human Movements Sciences, Medical Natural Sciences, Medicine, or Veterinary Medicine are not allowed to follow the course Anatomy & Physiology [BM41055]. They have to choose another course instead.

BM41035	Biomaterials	4
Responsible Instructor	Dr.ir. I. Apachitei	
Instructor	Dr.ir. E.L. Fratila-Apachitei	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	In improving patient care, especially in relation with tissue loss or dysfunction, the use of proper biomaterials plays a key role. The shift from tissue removal to tissue replacement and at present, tissue regeneration is driven by the (i)evolution of biomaterials from bioinert to bioactive and bioresorbable and (ii) the increasingly complex biomedical problems of an aging and more active population. As an example, the existing total joint replacements have a 75 - 85 % survivability at 15 years. However, since they were first introduced in the 1960s, average life expectancy increased by at least ten years impacting on the quality of the bone and implants survivability that deteriorate with age. As a consequence, improved implant survivability by 10 - 20 years and alternative treatment methods that delay or eliminate the use of prostheses are badly needed. The Biomaterials course covers in an interdisciplinary approach (biomaterial engineering and biology) the background information on type, composition, processing and properties of biomaterials along with their evolution. Metallic, polymeric and ceramic materials are included and their properties are described and compared with those of biological materials. The main methodologies for biomaterials characterization are covered as well as the procedures for biomaterials standardization and regulation. Further, the interactions between biomaterials and specific biological environments are explained and used for defining the specific requirements and selection principles for biomaterials. Here, biomaterials and medical devices for orthopedics, cardiovascular system and tissue engineering are covered. Finally, the research directions and approaches for development of novel biomaterials are outlined.	
Study Goals	At the end of the course the students will be able to: 1. Identify biomaterials used in: orthopedics, cardiovascular implants, and scaffolds for tissue engineering 2. Define the properties of biomaterials and their property map (range of property values) 3. Describe structure-property relationships for various biomaterials 4. Describe the methodologies for biomaterials characterization and regulation 5. Compare and contrast biomaterials and biological materials 6. Explain the main interactions between a biomaterial and a biological environment 7. Describe the principles for biomaterials selection and design 8. Describe approaches and processes used for functionalization of biomaterials 9. Identify, analyse and describe a biomaterial problem associated with a medical device 10. Explain the clinical performance of biomaterials and medical devices	
Education Method	Lectures, case studies and workshops	
Literature and Study Materials	Lecture slides and recommended literature (reader)	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

BM41050	Applied Experimental Methods: Medical Instruments	4
Responsible Instructor	Prof.dr. J.J. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	In this course, students carry out experimental research in instrument-tissue interactions within the context of application of minimally invasive techniques such as needle or catheter placement. Key in this course is that students familiarize themselves with making an experimental design, setting up an experiment and investigate the response to your experimental manipulations. The results should allow you to critically evaluate the chosen experiment parameters and data analysis methods.	
Study Goals	The student must be able to: Select and apply appropriate research methods. Create an experimental platform. Use suitable methods to analyse the data. Present the results in a concise report.	
Education Method	Students work in groups of two. They acquire basic knowledge through self-study and hands-on experience in experimental work.	
Assessment	Written report (at most 4 pages, two-column scientific paper format).	
Department	3mE Department Biomechanical Engineering	

BM41095	Medical Devices: Past, Present, Future	3
Responsible Instructor	Dr. D. Dodou	
Instructor	Prof.dr. J. Dankelman	
Instructor	Prof.dr. J.J. van den Dobbelaan	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. History of surgery 2. Minimally invasive techniques 3. Single-port procedures & NOTES 4. Percutaneous interventions 5. Robotic surgery 6. In vivo robots 7. Medical safety 8. Maintenance of surgical instruments	
Study Goals	The student will be able to describe the function, effectiveness, and challenges of surgical instruments in minimally invasive procedures. More specifically, the student will be able to: 1. illustrate advantages and disadvantages of minimally invasive surgery. 1. illustrate advantages and disadvantages of single-port procedures and NOTES. 3. describe instruments used during minimally invasive surgery, single-port procedures, and NOTES. 4. formulate criteria that surgical instruments have to fulfill in order to prevent sterilization problems. 5. indicate problems with quality control of surgical instruments. 6. describe the fields where robotic surgery is used and indicate the possibilities and limitations of the use of robotics. 7. classify technologies developed to train surgeons outside the operating theater. 8. discuss safety issues in surgery. 9. select information from the literature related to a certain issue and evaluate new developments.	
Education Method	Lectures. 4 hours per week	
Assessment	Homework assignments	
Department	3mE Department Biomechanical Engineering	

BM41155	3D Printing	4
Responsible Instructor	Dr. J. Zhou	
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Instructor	Ing. M.A. Leeftang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	in Track I (Musculoskeletal Biomechanics)and Track II (Medical Devices and Bioelectronics) of MSc Biomedical Engineering BME) and in the specialization of Bio-Inspired Technology (ME-BMD-BITE)	
Expected prior knowledge	BSc in mechanical engineering, materials engineering, industrial design engineering and aerospace engineering with a basic knowledge of materials (e.g. WB2330)	
Parts	- Overview of AM technologies, their principles and applications - Materials selection (polymers, ceramics, metals and composites) - Powder-bed fusion processes - Powdered materials and powder fabrication - Powder characteristics in relation to AM - Structural integration during melting and sintering - Extrusion-based systems - Design for functionality and 3D printability Software issues and AM limitations - Other powder-based manufacturing technologies as alternatives to AM - Post-processing of AM products - Mechanical testing of AM products - Case studies AM for engineering and medical applications - Assignment and presentations	
Course Contents	Additive manufacturing (AM), also known as 3D printing the process of joining materials to make objects from 3D model data, usually layer upon layer, as opposed to conventional subtractive manufacturing technologies, is set to revolutionize manufacturing across a wide range of industries. The course 3D printing - AM technologies is designed to equip students with the fundamental knowledge of AM and basic hands-on experience in AM. The topics include various AM processes, their principles, capabilities and limitations, materials used for AM processes, powdered material fabrication and characterization, CAD designs for AM, alternative powder-based manufacturing technologies, current and future AM applications, as well as the mechanical evaluation of AM parts. The primary focus is placed on powder-based AM for metallic parts. In addition to a series of lectures, project groups, each composed of 3-5 students, will create their own designs of the objects made of polymer PLA, formulate requirements in terms of geometry, dimensions, tolerances, mass, surface finish, functionality and mechanical properties of these objects, redesign them by making use of SolidWorks, print them, test and evaluate 3D printed products against the requirements, and document the projects in the form of reports and presentations. The new design is expected to reflect the distinctive features and freedom that AM has to offer (thin walls, lattice structures, form-free channels, integrated parts/functions, and customization), but within the capacity of the printing machines (bounding box size, minimum feature size or wall thickness). The report and presentation are expected to contain: reason for the selection of the object, weakness of the existing design, new design features, actual testing results and recommendations.	
Study Goals	Students will be familiar with the concept of AM, various AM processes, AM applications, materials for AM, product design for AM, post-processing techniques and evaluation methods. In addition, students will gain basic hands-on experience in product development via AM. Once successfully completing the course, the students will: - Have a fundamental knowledge of 3D printing/AM technologies for a wide range of applications - Be able to explain the benefits, advantages, challenges and limitations of individual AM technologies as well as alternative powder-based manufacturing technologies - Have an insight into selective laser sintering and melting (SLS/SLM) processes regarding process control in relation to the geometry and mechanical properties of the products - Become familiar with the common (metallic) materials for 3D printing and their characterization techniques - Understand printable geometry design rules and the common methods of digital file creation - Become familiar with the common techniques for post-processing - Become familiar with the common methods for the mechanical evaluation of 3D printed materials and functional parts.	
Education Method	Lectures and student projects	
Computer Use	Attending lectures and performing assignment are both compulsory	
Computer Use	- Object design with Software (SolidWorks Student Engineering Kit)	
Course Relations	Biomaterials, Tissue biomechanics and other engineering courses in MSc biomedical engineering, materials science and engineering, mechanical engineering, marine technology, aerospace engineering, design for interaction, integrated design and strategic design	
Literature and Study Materials	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21) - Lecture materials	
Books	- Additive Manufacturing Technologies, Ian Gibson, David Rosen, Brent Stucker and Mahyar Khorasani, 3rd Edition, 2021, Springer (chapters: 1, 2, 3, 5, 6, 7, 10, 15, 16, 18, 19 and 21)	
Reader	- Lecture materials	
Assessment	Written exam	
Department	3mE Department Biomechanical Engineering	

EE4555	Active Implantable Biomedical Microsystems	5
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Responsible Instructor	Dr.ir. C. Strydis	
Responsible Instructor	Dr. V. Giagka	
Responsible Instructor	T.M. Lopes Marta da Costa	
Responsible Instructor	Dr. D.G. Muratore	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none none	
Course Language	English	
Required for	- MSc Biomedical Engineering, Medical Devices track; - MSc Electrical Engineering, Biomedical Devices profile; - MSc thesis projects in the Section Bioelectronics.	
Expected prior knowledge	It is recommended but not required to have completed/attended the following courses: 1. Anatomy and Physiology (BM41055) 2. Fundamentals of Biomedical Engineering (BM41140) 3. Bioelectricity (ET4130) 4. Themes in Biomedical Electronics (ET4127) 5. Sensors and actuators (ET4257) 6. Measurement and Instrumentation (EE4C08) 7. Systems Engineering (EE4C02)	
Course Contents	Active Implantable Biomedical Microsystems (AIBM) in the human body are now becoming more widely acceptable and available since the development of cardiac pacemakers. Today, miniaturized wireless implantable systems are changing the face of biomedical research and clinical practices through the development of cochlear implants, neuroprostheses, braincomputer interfaces, deep organ pressure sensors, precise drug delivery units, bioelectronic medicine and electroceuticals. With a greater life expectancy and an increasing demand for medical healthcare to guarantee a sufficient health-related quality of life, there is a greater demand on technology and electrical and biomedical engineers to develop active implantable systems for a wide variety of medical diagnostics, treatments and therapies. This course provides a background on the application of AIBMs, an introduction to the latest techniques used and examples of existing medical devices. It can be used as a guide to the design of AIBMs and also as a reference for existing medical devices. The course is aimed at MSc students in Electrical Engineering (tracks Microelectronics, and Signals&Systems) and in Biomedical Engineering (track Medical Devices) who have an interest in the research and design of AIBMs. The following topics are treated: 1. Electrophysiology 2. Electrodes, optodes, sensors and actuators 3. Electrode characterization and modeling 4. Electronic system design 5. Stimulation electronics 6. Front-end electronics (signal conditioning, amplifiers, filters, ADCs, DACs, etc.) 7. Wireless data transfer 8. Energy harvesting and power delivery 9. Applications: cochlear implants and cardiac pacemakers 10. Digital design 11. Biocompatibility, materials and fabrication considerations 12. Assembly, system integration and packaging 13. Application: spinal cord stimulator 14. Reliability and security 15. Risk management and/or 16. Ethics	
Study Goals	After successful completion of the course, you: 1. Know, understand and can apply (3) the functionality, construction and application of active implantable biomedical microsystems (AIBMs) and their sub-systems and parts; 2. Know, understand and can apply (3) the fundamental principles and approaches for designing an implantable system; 3. Know, understand, can apply, analyse and can reason about (evaluate; 5) the design requirements, application-specific implementation challenges, and appropriate design choices of existing AIBMs, incl. cardiac pacemakers, cochlear implants and spinal cord stimulators; 4. Know and understand (2) the most important implantability, biocompatibility, reliability, safety and security issues of AIBMs; 5. Know and understand (2) the most important regulatory and clinical constraints of AIBMs 6. Can reason (5) about future developments of AIBMs, a.o. electroceuticals and bioelectronic medicine; NB. The numbers correspond with Bloom's taxonomy.	
Education Method	1. Lectures 2. Self study 3. Case study 4. Group project	
Literature and Study Materials	1. a collection of book chapters, thesis chapters and review/overview papers; 2. handouts of the slides used during the lectures, 3. video recordings of the lectures all made available via the online learning system (Brightspace). 4. scientific papers (IEEE Xplore, PubMed); 5. patents; 6. user manuals; 7. patient manuals; 8. manufacturer websites; 9. the Internet in general, to be consulted by the students themselves.	
Assessment	The student has to: - Form a group (6 students/group) in the first week, based on affinity with the topics below. - As a group, perform a case study of a complete device, which will be announced in Wk. 4.2.	

- Investigate 6 topics:
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- Prepare an annotated slide set on the first 3 topics in Wk. 4.5
 - *Electrodes and electrophysiology
 - *Stimulation and power management electronics
 - *Recording electronics and wireless data transfer
 - *Max. 12 slides (4 per group of 2 students), using a prescribed template
 - *To be uploaded on Brightspace before Wk 4.6
 - *To be commented on in written form and handed in by another group during the 1st lecture in Wk. 4.7
 - *Sub-group (of 2 students) on topic A comment on topic B; B -> C; C -> A
- Prepare a group presentation on the second 3 topics:
 - *Analog, mixed and digital system design
 - *Microfabrication, system integration, packaging, implantation and biocompatibility
 - *Reliability, safety and security; clinical and regulatory considerations
- One student, to be randomly selected by the teachers on the day of the presentation, will do the presentation (max. 12 minutes) for the entire group in Wk. 4.10 or 4.11.
- Other groups will discuss the presentations (4 minutes) and ask questions and will comment briefly (max. 2 minutes per group / presentation) on all presentations

The teachers will:

- Organize/manage the course
- Give the lectures
- Provide relevant material and background knowledge
- Define the case study
- Give the templates for the annotated slides and the comments
- Grade the slide sets per group of 2 students
- Grade the comments on the slide sets per group of 2 students
- Return the commented slide sets to the authors
- Buy the T-shirts to wear during the final presentations
- Grade the presentations per group of 6 students
- Grade the comments made on the presentations per group of 6 students
- Decide on the final marks

The final mark is based on the grade received for the annotated and commented slide sets (37.5% + 12.5% (bonus/malus), respectively), and the grade received for the group presentation (45%) and the comments made on another group presentation (5%).

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).

ET4127	Themes in Biomedical Electronics	4
Responsible Instructor	Prof.dr. P.J. French	
Responsible Instructor	Prof.dr.ir. W.A. Serdijn	
Gast Instructor	M. Mastrangeli	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Bachelor in Engineering, preferably Biomedical Engineering or Electrical Engineering.	
Course Contents	The course addresses the impact of microtechnologies on medicine and biology. The following topics are part of the course: - Biomicroelectromechanical systems (BioMEMS) - Sensors & Actuators - Micro- and nano-fluidics - Lab-on-CMOS - Electrophysiological recording systems - Neurostimulation - Electroceuticals and bio-electronic medicine - Electromagnetics for MRI - Ultrasound (incl. intravascular ultrasound) - Biomedical signal processing - Biocompatibility and safety	
Study Goals	Introduction to microtechnologies in medical and biological applications. Upon successful completion of the course the student knows and understands the role and basic workings of BioMEMS, biosensors, biomedical circuits and systems, microfabrication for biomedical devices, biomedical signal processing microfluidics, bioelectronics, ultrasound, photo-electronics in monitoring, diagnosis and treatment.	
Education Method	Lectures, self-study, essay (mini-thesis)	
Literature and Study Materials	Study material will be posted on Brightspace and is available from the (guest) lecturers on request.	
Assessment	Minithesis on a topic of the course and (depending on lecture attendance) homework (25%). The minithesis has the form of a scientific article (6-10 pages, figures included, double-column format) and offers a review of the topic of study and the author's own interpretation/extrapolation/view on the matter. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-MD Obligatory electives: Select 3 courses from the following list

BM41075	Regenerative Medicine	4
Responsible Instructor	Dr.ir. E.L. Fratila-Apachitei	
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Through regeneration, a living system can restore some of its lost cells, tissue or organs. The fascinating field of regenerative medicine is harnessing the regeneration principles for treatment of human diseases. How to induce and/or control the regeneration process in the adult system is one of the fundamental challenges addressed in this field. After introducing the main concepts and components, the course will present the achievements of regenerative medicine for selected tissues/organs based on a systematic approach including: the pathology involved, the regenerative therapies at molecular, cellular, tissue and/or organ level, and the state-of-the-art results. For each pathology, the following aspects will be discussed: (i) specific requirements for regenerative therapies (e.g., cells type and source, soluble factors, biomaterials/scaffolds), (ii) enabling technologies for cells manipulation, generation and evaluation of scaffolds/tissue/organ, (e.g., organs-on-chip, micro/nanofabrication, bioprinting), (iii) the influence of cells and soluble factors, biomaterials/scaffold properties, and host conditions on the regeneration process, (iv) the relevant cellular and molecular interactions, (v) biocompatibility assessment of regenerative systems and (vi) future challenges for such treatments.	
Study Goals	At the end of the course, the students should be able to: 1. Describe the type of cells, soluble factors, biomaterials and engineering processes used in regenerative medicine 2. Describe and compare various regenerative approaches for a certain pathology 3. Discuss the effects of cell type, biomaterials/scaffold properties and host microenvironment on regeneration processes 4. Identify the main cellular signaling pathways involved in a certain regenerative process 5. Describe models and methods for biocompatibility assessment of a specific regenerative system 6. Propose, justify and describe regenerative approaches/products for a given pathology 7. Recognize the clinical, regulatory and ethical issues of regenerative therapies	
Education Method	Blended education methods including lectures and group projects involving specialists from university medical centers and TU Delft.	
Literature and Study Materials	Lecture slides and selected literature for projects and further reading.	
Assessment	Written exam and project.	
Department	3mE Department Biomechanical Engineering	

BM41090	Computational Mechanics of Tissues and Cells	6
Responsible Instructor	Prof.dr. A.A. Zadpoor	
Instructor	N. Tümer	
Instructor	M.J. Mirzaali Mazandarani	
Contact Hours / Week x/x/x/x	0/0/3/3	
Education Period	3 4	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Required for	This course is required for students following the Neuromusculoskeletal Biomechanics track of the Biomedical Engineering master programme. Other students are more than welcome to enroll.	
Course Contents	The course consists of two parts: lectures and hands-on workshops. The lectures are designed to give students in-depth knowledge of the mechanical behaviour of cells and tissues and how their behaviour can be explained in terms of mechanical theories. The hands-on workshops give students the opportunity to apply what they have learned in the lectures. A commercial Finite Element (FEM) modelling package is used to make FEM modelling as practical as possible. The emphasis is therefore shifted from the technical details of the FEM modelling towards understanding the modelling procedures and applying them for analysis of certain problems. The problems are chosen to be as close to the real-world problems as possible. In addition to finite element modelling, the principles of medical image processing for finite element modelling are discussed. The course is divided into two almost equal parts. The first part covers patient-specific finite element modelling of tissues and implants. The second part is focused on the mechanical behaviour of cells and proteins (such as collagen). The following topics will be covered in the lecturing (tentative): Lecture 1: Introduction to the course Lecture 2: Geometry of tissues and implants for finite element modelling (including statistical shape and statistical shape and intensity models) Lecture 3-5: Material modelling and material properties (including isotropic and anisotropic linear elasticity, hyperelasticity, viscoelasticity, and poroelasticity) Lecture 6: Loads and boundary conditions (including musculoskeletal models) Lecture 7: Modelling of tissue growth and adaptation Lecture 8: Introduction to cell mechanics, continuum models of the cell Lecture 9: Tensegrity models of the cell Lecture 10: Polymer-based models of the cell Lecture 11-12: Protein mechanics The workshops start from very simple modelling exercises that introduce the FEM package and show how it can be used for simple modelling tasks. The level of difficulty will gradually increase up to the point that students can do FEM modelling of tissues, cells, proteins, and tissue-implant systems.	
Study Goals	Students who have successfully completed this course must be able to: 1.Explain continuum theories that are used for modelling of biological tissues (i.e.anisotropic elasticity, viscoelasticity, and poroelasticity) 2.Translate physiological/pathological conditions of cells and tissues to the language of mechanics 3.Comfortably use a certain commercial FEM package (possess general FEM modelling skills) 4.Build FEM models of tissues and cells 5.Identify different modelling approaches 6.Make appropriate choices concerning material models, boundary conditions, loadings, etc. 7.Critically assess the results of their FEM models	
Education Method	(Interactive) lectures and hands-on workshops	
Computer Use	Computer will be used for the hands-on workshops	
Literature and Study Materials	Recommended texts for the course: 1.Cowin, Stephen C., Doty, Stephen B., 2007, Tissue Mechanics, Springer. 2.Cowin, Stephen C. (Editor), 2001, Bone Mechanics Handbook 2nd Ed., CRC Press. 3.Bartel, Donald L., Davy, Dwight T., Keaveny, Tony M., 2006, Orthopaedic Biomechanics: Mechanics and Design in Musculoskeletal Systems, Prentice Hall. 4.Mow, van C., Huiskes, Rik, 2004, Basic Orthopaedic Biomechanics & Mechanobiology 3rd Ed., Lippincott Williams & Wilkins. 5.Jacobs, Christopher R. et al., 2012, Introduction to Cell Mechanics and Mechanobiology, Garland Science.	
Prerequisites	Students should have working knowledge of the mechanics of materials	
Assessment	Final exam (for the lecturing part) + submission of lecture (individual) and workshop (group) assignments: The grading policy for this course includes: Workshop assignments (total: 60%, first two assignments: 15% each, last workshop assignment: 30%). Lecture assignments (total: 15%, each assignment: 5%) Written exam (25%) 50% of the grade of the written exam is needed to pass the course. At least 2 out of 3 lecture assignments have to be submitted within the deadline provided in the course to pass the course The three workshop assignments are team work activities, and the final report has to be approved by all group members. Once the group members are defined, they cannot be changed. In case of any problems with your group members, we preserve the right to ask for an oral examination. If students fail the re-sit exam, in order to pass the exam in the next academic year, they are asked to not only retake the exam but also submit again two lecture assignments. Any delays with the submission of the lecture assignments will result in down grading of your grade. No delays in the submission of workshop assignment is accepted.	
Department	3mE Department Biomechanical Engineering	

BM41100	Medical Devices: Designing for CE-marking	3
Responsible Instructor	A.J. Loeve	
Co-responsible for assignments	R. Straathof	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The assignments follow the line of the lectures (see course syllabus) and have to be completed using one of a selection of actual prototypes that were manufactured within the MISIT group. The assignments all tackle a part of the CE-marking process, such as risk analyses, redesign for cleanability, materials selection etc.(detailed info in the course syllabus on Brightspace).	
Study Goals	The assignments and final reports will be judged on: - Demonstration of concise line of reasoning and critical thinking - Effort to be complete - Originality of approach and proposed redesign, and contribution to minimize risks - Use of information sources - Written presentation The final presentation will be judged on: - Balance between main points and details - Liveliness - Clear message - Visuals - One's own reflection on the course and learned aspects	
Education Method	Each week, a different aspect is discussed related to the eventual goal of receiving permission to use a medical device on patients in the hospital. The lectures (one hour each week) are given by the course lecturers and guest lecturers affiliated to hospitals, research institutes and companies. After each lecture, there is some time for discussion and questions about the previously made assignment. The assignments are completed by teams of four (sometimes five, depending on the class size) students working in close collaboration. The final report consists of all assignments having been put together after processing the feedback. On the last course day, the student teams will have to present their final results and grade each other.	
Assessment	Final presentation (20%), assignments grades (40%) and final assignment report (40%)	
Remarks	Due to the extensive feedback cycles in this course, no more than 30 students will be admitted. The application and admission procedure will be communicated via the course's Brightspace page as soon as it opens (usually about 2 months before the start of the course).	
Department	3mE Department Biomechanical Engineering	

BM41131	Tissue Biomechanics	3
Responsible Instructor	A.C. Akyildiz	
Instructor	Dr.ir. M.P. Peirlinck	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course involves lectures and interactive group discussions. In the lectures, the students will be introduced to the biomechanics of various tissue types in the human body, including bone, cartilage, tendon, ligament, muscle, and vessel tissues. The students will be guided through the topics of these tissue types focusing on structure-function relationship, material behavior (experimental examples), combined with modeling approaches and clinical application. The fundamental knowledge on continuum mechanics, required for a good understanding in tissue biomechanics, will be provided as well. Additionally, experts will be invited to talk about their current scientific research on tissue biomechanics, representing the rapidly developing field of tissue biomechanics. During the group discussions, the students will be expected to read and take a deeper dive into selected tissue-specific research articles and discuss the topics background, approach, methods, results, and limitations with their peers. Active participation in these group discussions is obligatory in order to take the final examination.	
Study Goals	After successful completion of this course, the student should be able to: 1. Explain the biomechanical constituents, hierarchical structure and mechanical function of biological tissues, and how they relate to each other, 2. Discuss the basic concepts of continuum mechanics that are relevant for tissue biomechanics, 3. Examine the mechanical behavior of biological tissues using continuum mechanics framework, 4. Evaluate recent scientific research on tissue biomechanical modeling and experimentation, and its applicability in the clinic.	
Education Method	Lectures, group discussions	
Literature and Study Materials	Book / online material recommended to complement lectures: An introduction to biomechanics: solids and fluids, analysis and design (Jay D Humphrey & Sherry L O'Rourke) Nonlinear solid mechanics: a continuum approach for engineering (Gerhard Holzapfel) Continuummechanics.org	
Assessment	Group assignments Written exams	
Department	3mE Department Biomechanical Engineering	

EE4109	Structured Electronic Design	5
Responsible Instructor	Dr.ir. C.J.M. Verhoeven	
Instructor	A.J.M. Montagne	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	A grade of a minimal 8 for this course is required to obtain access to the masterclass EE8011MSC	
Expected prior knowledge	This course required the knowledge and skills of MSc course EE4C10 and BSc course EE3C11. This course uses the same book used for BSc course EE3C11. A free download of the book is available via: https://analog-electronics.tudelft.nl Chapters that contain prior knowledge are: 1, 2, 7,8, 9,10 (excluding 10.3.5, 10.3.7, 10.3.8) , 11 (excluding 11.4.4, 11.4.6, 11.4.7, 11.5.1) and 12.1 and 12.2 Students that did not follow BSc course EE3C11 should (could) follow the homologation program from: https://analog-electronics.tudelft.nl/Homologation/courseWebSite/index.html Here you can find lecture material and videos that cover all electronics topics of EE3C11. The teaching staff can assist in studying the homologation program. On request, the lecturers can provide extra lectures for external/international students on topics treated in EE3C11 on background knowledge, like poles and zeros.	
Course Contents	This course deals with the structured design of analog electronic circuits using CMOS(and Bipolar) transistors. The design method presented in this course is based on network theory, control theory, signal processing, and physics. Many people consider the design of analog electronic circuits to be complex. This is mainly because designers have to deal with many relevant performance aspects that can be achieved in many different ways. In other words, there are many degrees of freedom for obtaining the desired performance of an electronic circuit. Theoretical concepts, circuit topologies, electronic devices, their operating conditions, and the physical layout of a system constitute an enormous design space in which it is easy to get lost. For this reason, analog electronics is often regarded as an art rather than a solid discipline. At first glance, there doesn't seem to be a straightforward way to design analog circuits and systems: experienced designers intuitively use all these degrees of freedom to modify and combine known solutions into new ones. However, intuition is knowledge, the origin of which has become unclear. It cannot easily be shared with novices, and therefore, it cannot be a basis for the education of designers. However, the design of electronic circuits can be shared and understood if it is presented in a structured way. This requires a precise formulation of design goals and strategies and a clear distinction between theoretical concepts and their physical implementations. Instead of taking numerous existing solutions to known problems as a starting point, it is far more effective to start a new design with clear distinction between theoretical concepts and study how they can be implemented in such a way that the possibilities of the implementation technology are maximally exploited. Such an approach results in well-defined and reliable designs with predictable performance and provides a basis for sharing and developing knowledge. During the engineering phase, designers need to find or create real-world components, determine their operating conditions, and fix them by design. To this end, they need to formulate and solve so-called design equations for the circuit. Design equations relate the properties and the operating conditions of the electronic devices to the performance requirements and the cost factors of the circuit. For example, to determine a budget for the gain-bandwidth product of an operational amplifier, we must find relations between this gain-bandwidth product and the performance requirements of the circuit that comprises this operational amplifier. I Setting up design equations for analog circuits, especially at transistor-level is a difficult and cumbersome task. It requires knowledge about technology, modeling techniques, and interactions between the behavior of the circuit and the transistor properties. It goes hand in hand with symbolic analysis. Performing symbolic calculations by hand is sensitive to errors, and in practice, only a limited number of circuit performance aspects can be considered. For this reason, we will use SLiCAP (Symbolic Linear Circuit Analysis Program) during this course. SLiCAP (https://analog-electronics.tudelft.nl/SLiCAP.html)	
Study Goals	The main goal of this course is to provide the student with the essential knowledge and skills to design an amplifier in BiCMOS or CMOS technology according to given specifications and to motivate and fully understand all design steps involved. After the course the student is able to: - work in a design team in a complete CMOS design - translate an application description of an amplifier into a specification comprising, a functional specification, specification of performance measures, operational conditions and cost factors. - design a proper amplifier concept based on source and load specification. - evaluate the amplifying capabilities of basic amplifier stages such as CS and CE stage for a given technology - identify performance aspects such as noise, signal handling capability and bandwidth of basic amplifier stages such as CS and CE stage for a given technology - scale devices and select their operating conditions to influence these performance aspects - to apply error reduction techniques, such as compensation, balancing and (local) negative feedback to improve the performance of these basic amplifying stages - to motivate the decision to use multi-stage amplifiers - to design (high-performance) multi-stage amplifiers using error reduction techniques such as compensation, balancing and feedback. - setup design considerations for noise and take adequate design decisions	

- setup design considerations for signal handling capability and power efficiency and take adequate design decisions
- setup design considerations for frequency response and take adequate design decisions
- setup design considerations for weak distortion and take adequate design decisions
- setup design considerations for biasing and design and implement proper biasing circuitry
- evaluate the performance versus requirements of the final design and propose options for improvement
- select proper models and design verification tools for each design step
- write a design report in which all design steps are motivated and the performance is evaluated and explained.

Education Method

Lectures and Colstructions:

The systematic electronics design method will be illustrated during the lectures and colstructions with a design example made in SLiCAP.

The design example closely relates to the hearing loop system designed during the BSc course EE3C11. This BSc design was based on operational amplifiers as controllers. In this course, the focus will be on the design of the controllers as transistor circuits.

During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design. The design steps are documented in a report using SLiCAP.

The design report can be efficiently used as a basis for the exam, which is organized as a design review for the design team.

Computer Use

A computer with SLiCAP and the free circuit simulator LT-Spice is necessary. Help for installing and using SLiCAP will be available during the lectures.

SLiCAP is an acronym for Symbolic Linear Circuit Analysis Program.

See: <https://analog-electronics.tudelft.nl/SLiCAP.html>

Literature and Study Materials

Book (also used during BSc course EE3C11):

Structured Electronic Design Edition downloadable from:

<https://analog-electronics.tudelft.nl>

Assessment

During the first two weeks of the course, the students are invited to start or join a design team. In these teams, the students work on a design related to the example design used in every lecture to illustrate the application of the theory from the lectures.

At the end of the lecture period, the student team makes an appointment review of their design. The design review consists of two sessions:

- 1) A design preview, where the students present and defend their design in a relaxed atmosphere. They can use the feedback to make improvements.
- 2) A design review, where the students present and defend their design again. Each team member has to give substantial input during the session. After this session, each student is graded individually.

The design report, created with SLiCAP during the lectures by the team, can be efficiently used as a basis for the review.

If there are logistic or personal reasons that make it impossible for a student to join a design team, it is possible to opt for an individual design review (oral exam). Please contact the educators about this.

In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024.

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EE4C08	Measurement and Instrumentation	5
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Instructor	T. Manzanque Garcia	
Practical Coordinator	M. Spirito	
Practical Coordinator	Ir. P.J. Aubry	
Practical Coordinator	Ing. J. Bastemeijer	
Practical Coordinator	F. Simjanoski	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of the following material is very important: - circuit analysis - feedback systems - sampling theory	
Course Contents	This course offers a broad introduction to measurement and instrumentation systems. The basic elements of such systems will be presented: transducers, amplifiers, filters and analog-to-digital converters. The different sources of error (noise, interference, offset, non-linearity, aliasing) that limit the performance of such systems will be discussed. Methods of analytically determining the corresponding detection limits will be presented, and in particular how errors can be referred to the input of a system and represented by equivalent voltage/current sources. Several techniques for mitigating measurement errors will be discussed in detail e.g. the use of feedback, filtering, synchronous modulation, chopping, auto-zeroing, dynamic element matching. The course will conclude with a measurement laboratory. Students may choose ONE of the following topics: Device characterization, RF measurements or LF measurements.	
Study Goals	Students should be able to: - understand the specifications of instruments and measurement systems - determine the main error sources and the associated detection limits of measurement systems - use measurement techniques to reduce measurement errors - perform basic RF/LF/DC measurements	
Education Method	Chapters: the course is divided into 10 chapters, which will be covered in weeks 1.1 - 1.6. Each chapter is divided into modules, which come with slides with pre-recorded explanations. Study these chapters before the corresponding tutorial, as indicated in the weekly modules on Brightspace. Tutorials: during these live interactive sessions you can test your understanding of the lecture material using exercises and quizzes (4 hr/week) Laboratory: 4 hrs/week (weeks 1.7 and 1.8) Details are provided on the Brightspace page of the course. Note that this information may change depending on the developments around Covid-19.	
Literature and Study Materials	Textbook: "Introduction to Instrumentation and Measurements," 3rd Edition, by Robert B. Northrop Handouts and slides provided via Brightspace Note: the following book was used in previous years and can still be used as an alternative textbook: "The Art of Electronics," 3rd edition, by Horowitz and Hill	
Assessment	The final grade will be built up as follows: Written exam = 60% Homework = 20% Practical assignment = 20% OR Written exam = 80% Practical assignment = 20% whichever of the two is higher. The written exam will take place on campus. It is an open-book exam with open questions. The type of questions will be similar to previous years (see example exams on Brightspace). The exam will be graded on a scale of 1-10. The homework grade is the average of the grades obtained for the 4 homework assignments, each of which will be graded on a scale of 1-10. The homework grade is valid for one year, i.e. the above calculation applies for the exam and the resit. The grade of the practical assignment is based on the lab report submitted at the end of the course. This will be graded on a scale of 1-10. If you re-take the course next year, your homework grade from this year will not carry over to next year. In that case, you can choose to do the homework of that year (and we encourage you to do so). Your lab grade does carry over to next year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	You are allowed to use a calculator, the book, and a print-out of the slides or other material that you might want to consult. The use of a smartphone, tablet, laptop or other electronic devices (except a calculator) is not allowed. See the Exams folder on Brightspace for example exams from last year. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.drs. Q. Fan	

EE4C11	Systems Engineering	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	R.T. Rajan	
Responsible Instructor	S. Izadkhast	
Instructor	Prof.dr. P.J. French	
Instructor	Dr.ir. C.J.M. Verhoeven	
Instructor	P.P. Sundaramoorthy	
Instructor	C. Galuzzi	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	The course is centered around the V-model commonly employed in systems engineering, in which a problem/project is gradually refined from high-level idea to detailed specification (followed by implementation and testing, which are not part of this course due to time limitations). The course includes 5 plenary instruction sessions focusing on subsequent steps in the V-model. The course takes a hands-on approach in which groups of about 6-8 students with different backgrounds work on a case study. Groups meet weekly with a TA to discuss progress and plan activities. In addition tutors from the teaching staff (case owners) will be available for assistance in problem formulation and formative feedback on deliverables. The course will be organized in 3 phases, the first two ending in formal reviews and the third phase concluding with a customer/investor pitch. The course concludes with a final symposium session in which all groups present their final (product) designs.	
Assessment	The final grade will be composed of a group grade based on - reviews (3x) - case report - documentation and an - individual oral assessment In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers. No resit will be offered; instead failing students will be assigned additional tasks to remedy any deficiency. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ET4130	Bioelectricity	3
Responsible Instructor	Dr. V. Giagka	
Instructor	Dr. V. Giagka	
Instructor	T.M. Lopes Marta da Costa	
Contact Hours / Week x/x/x/x	0/0/3/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BM41065 (Medical Technology I Diagnostic Devices & Healthcare Systems) (not compulsory) Basic knowledge of electricity	
Course Contents	Course main goal: This course aims to give students a basic understanding of bioelectric phenomena (like the Action Potential, the transmission of neural signals and the phenomena taking place during electrical stimulation of neurons) and their mathematical analysis, based on biophysical/mathematical source models and volume conduction theory. Course description: In this course the electrophysiological and biophysical processes underlying the generation of bioelectrical activity are presented in relation to the potential differences that can be measured on (or in) the human body. These electrical signals may provide information on the (patho)physiological condition of the corresponding organ. Also, the principles and practicalities of artificial neural stimulation are dealt with. As a general introduction the theory of volume conduction of ionic currents, based on the quasi-static expression of Maxwell's equations, is presented. This theoretical framework is then applied to nerves and neurons. The aim is to obtain a basic understanding of the electrical phenomena involved, emphasizing both on a quantitative description and a model-oriented approach. This knowledge will serve as well as a basis for understanding the basics of neuromodulation (electrical, optical and ultrasound), and clinical measurement methods of ENG, EEG and ECG. The course also serves as an introduction to the extensive international scientific literature on bioelectric phenomena.	
Study Goals	Upon successful completion of this course, the student will be able to: 1. Describe the important experiments and findings (discussed in this course) related to the field of bioelectromagnetism. 2. Explain and sketch the anticipated behaviour of impulse propagation, and argue about the factors that affect it. 3. Describe and explain the subthreshold membrane phenomena. Draw the models that describe the behaviour, derive related equations, and solve them to calculate equilibrium conditions and characteristics. 4. Describe and explain the active behaviour of the membrane. Explain the models that describe the behaviour, and the experimental methods that can be used to quantify it. Relate the methods to their measurable components. 5. Describe and explain the bioelectrical measurement methods and models and compare them. 6. Describe and explain the processes taking place during electrical, optical and ultrasound stimulation of neurons, and argue about the stimulation methods used.	
Education Method	Lectures, self-study	
Literature and Study Materials	1. Book: Bioelectromagnetism (by T.Malmivuo and R.Plonsey), electronically available at www.bem.fi. 2. Lecture notes.	
Assessment	Written and/or online examination In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4257	Sensors and Actuators	4
Responsible Instructor	Prof.dr. P.J. French	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	P-study	
Course Contents	The course silicon sensors and Actuators gives an overview of the most important principles related to sensors and actuators fabricated in integrated silicon technology. The sensors are divided into those for optical, mechanical, thermal, magnetic and chemical signals. These domains will be dealt with from basic principles leading to the applications. The second part of the course will deal with actuators. The actuators lectures give the range from large machines down to silicon micromachined device in the micron range. The course is designed for students who will perform their thesis work in one of the laboratories within the faculty working on or using sensors	
Study Goals	The aim of this course is to learn about the physics and electronics of transducers. This brings together different disciplines develop these systems.	
Education Method	Lectures	
Literature and Study Materials	Lecture notes Sensors & Actuators	
Assessment	Written, essay or oral. Assessment material: at least 6 chapters from the book Sensors & Actuators. Students can choose which form of assessment they want to take. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

ET4386	Estimation and Detection	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	R.T. Rajan	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	The course gives a solid background for EE4715 Array Processing and EE4685 Machine Learning, a Bayesian Perspective.	
Expected prior knowledge	A course on random signals or stochastic processes, like EE2S31.	
Course Contents	This course covers the basics of detection and estimation theory, as used in statistical signal processing, adaptive beamforming, radar, telecommunication, system identification, and elsewhere. Part I: Optimal estimation covers minimum variance unbiased estimators, the Cramer-Rao bound, best linear unbiased estimators, maximum likelihood estimation, recursive least squares, Bayesian estimation techniques, and the Wiener filter.. Part II: Detection theory covers simple and multiple hypothesis testing, the Neyman-Pearson Theorem, Bayes Risk, and testing with unknown signal and noise parameters.	
Study Goals	- To be able to explain some key problems in detection and estimation. - To be able to explain the main tools to solve these problems. - To be able to implement these techniques in Matlab. - To be able to apply these techniques to new detection and estimation problems.	
Education Method	Lectures plus project.	
Computer Use	Access to Matlab.	
Literature and Study Materials	- Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory; S.M. Kay, Prentice Hall 1993; ISBN-13: 978-0133457117. - Fundamentals of Statistical Signal Processing, Volume II: Detection Theory; S.M. Kay, Prentice 1993; ISBN-13: 978-0135041352.	
Assessment	The exam for et4386 Estimation and Detection Theory will be a written exam. The exam is closed book, but, students are allowed to bring a double sided self HANDWRITTEN A4 formula sheet. As part of the course, there is a compulsory mini project, which helps you to get experienced with the theory and to apply this to a practice problem. The available mini projects will be announced via the course website, after which students can sign in. The mini projects can be done in pairs. The final report on the project needs to be submitted in pdf before the date announced on the website. The project report will be graded and counts for 20 % of the final grade. Access to the final exam will only be granted if the project is handed in. In case of an insufficient final result, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

ME41065	System Identification and Parameter Estimation	7
Responsible Instructor	M.L. van de Ruit	
Instructor	Prof.dr.ir. A.C. Schouten	
Instructor	Dr.ir. W. Mugge	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic control engineering and experience with Matlab	
Course Contents	System identification deals with the estimation of physical properties of (dynamic) systems from measured data (inputs and outputs), which is essential for the understanding of system behavior. System identification is particularly useful for biological systems where physical entities are generally not known a priori. The course provides mathematical a background of system identification in both time domain and frequency domain with a strong focus on practical application. Major topics to be studied are: spectral densities, nonparametric black-box models, discrete time domain models (parametric grey-box), closed loop systems (utilizing feedback), translation of transfer functions into physical parameters, optimization routines and design of perturbation signals. During the course examples from both technical and physiological systems (i.e. estimate the behavior of a driver or the dynamics of a human joint) will be discussed.	
Study Goals	At the end of the course the student is able to. LO1: Signals a: explain the importance of input signal design and signal acquisition properties for the identification of system dynamics. b: select optimal input signals to identify unknown system dynamics and test them using MATLAB and Simulink. LO2: Systems a: illustrate how system dynamics can be accurately estimated using non-parametric (non)linear time-invariant models in time- and frequency domain. b: estimate nonparametric models of unknown systems from measured input and output signals using MATLAB and Simulink. LO3: Models a: explain what model structure is best fits a non-parametric estimate and which corresponding optimization algorithm to use to find the model parameters. b: parameterize physical models from nonparametric estimates employing optimization techniques using MATLAB and Simulink.	
Education Method	Self study, lectures (2x2 hours per week) including (home) work assignments	
Prerequisites	recommended prior knowledge - Probability theory and statistics (e.g. WBMT2049-T1) - Signal analysis (e.g. WB2235) - Systems and control engineering (e.g. WB3240)	
Assessment	Final closed book exam (60%), digital midterm (40%) tests and assignments (pass/fail)	
Tags	Analysis Mathematics Matlab Mechanics Modelling Numeric Methods Programming Signals Signals and Systems Small groups Technology	
Department	3mE Department Biomechanical Engineering	
Contact	By email SIPE-3ME@tudelft.nl	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-MP Track Medical PhysicsMEDICAL PHYSICS (MP)

Introduction 1	The Medical Physics track especially focuses on techniques from medical imaging, medical image analysis, scientific visualization and radiotherapy. In medical imaging methods are studied such as Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and Nuclear Imaging, that all provide high-quality 3D and 4D information of the human anatomy and function. Furthermore, medical image analysis methodology is explored for processing and understanding the medical imaging data. In radiotherapy, clinical, technical and bio-physical concepts resulting in optimised treatment planning are addressed. Additional topics involve methodology from medical physics, such as related to radiological health issues, machine learning and nuclear chemistry.
Exit Qualifications	FINAL TRACK QUALIFICATIONS MEDICAL PHYSICS A graduate in Biomedical Engineering Medical Physics is able to: 1. Competent in the scientific discipline Biomedical Engineering Medical Physics 1a. apply a broad and profound knowledge of physical methods to biomedical engineering problems. 1b. apply general and more advanced but specialized knowledge of medical imaging, radiological health physics, image processing and radiotherapy to selected biomedical engineering problems. 1c. apply knowledge of anatomy and physiology, experimental design and statistics, to problems from the medical physics specialization/focus area. 1d. apply advanced methodologies of at least two subjects from the fields of medical visualization, nuclear medicine, optical imaging and machine learning to biomedical engineering problems. 2. Competent in doing research 2a. independently generate new scientific knowledge and new insights within the field of medical physics. 2b. assess research, including scientific literature, in relation to problems from the field of medical physics on its scientific value. 3. Competent in designing 3a. implement and evaluate physical solutions by designing appropriate experiments to do so. 3b. generate innovative contributions to the discipline of medical physics. 4. A scientific approach See general BME Final qualifications 5. Basic intellectual skills See general BME Final qualifications 6. Competent in operating and communicating See general BME Final qualifications 7. Considering the temporal and social context See general BME Final qualifications

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-MP Obligatory courses (24 ECTS)

Introduction 1

Please note that students that have a background in Biology, Biomedical Engineering/Technology, Biomedical Sciences, Biopharmacy, Clinical Technology, Human Movements Sciences, Medical Natural Sciences, Medicine, or Veterinary Medicine are not allowed to follow the course Anatomy & Physiology [BM41055]. They have to choose another course instead.

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morphological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 6 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: $((0.85 \cdot EX + 0.15) \cdot AS) / 2$ or without the bonus point from the practicals: $(EX + AS) / 2$ With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

AP3232	Medical Imaging Signals and Systems	6
Responsible Instructor	Dr.ir. M.C. Goorden	
Instructor	Dr. D. Maresca	
Instructor	Dr. S.D. Weingärtner	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	A thorough mathematical basis is essential to follow the class. This can for example include Linear Algebra I + II and Calculus I + II. An in-depth understanding of Fourier analysis and basic physical understanding of electromagnetism is necessary."	
Course Contents	The course Medical Imaging Signals and Systems covers the principles of advanced medical imaging modalities. Particularly, students will study the physics, acquisition techniques and signal processing underlying conventional X-ray, Computerized Tomography (CT), Single Photon Emission Computed Tomography (SPECT), Positron Emission Tomography (PET), Ultrasound (US) and Magnetic Resonance Imaging (MRI). Course Content: Introductory lecture with the history of medical imaging and its importance for modern medicine and biomedical research. Conventional X-ray, CT, SPECT and PET imaging (33% of content): Basic principles of planar X-ray imaging and scintigraphy (short); Advanced methods for image acquisition using CT, SPECT and PET; Mathematical techniques for image reconstruction, including filtered backprojection and iterative approaches; Issues of image quality, especially related to resolution, signal to noise ratio, quantitative and dynamic imaging. MR imaging (33% of content): Fundamentals of MRI: spin, Larmor frequency, T1, T2, T2* contrast (short); Spatial encoding: K-space; Image Quality: PSF, signal strength and noise level; Advanced MRI techniques such as diffusion, perfusion and flow imaging. Ultrasound imaging (33% of content): Physical and mathematical principles of ultrasound imaging, including probe design and advanced US imaging techniques such as harmonic imaging and non-linear reconstruction techniques for imaging tissue properties. The course will end with a mini-symposium on specialized imaging techniques. Therefore, recently developed medical imaging methods will be studied, going beyond the theory of the lectures. Students working in pairs will write a short essay on a selected technique and present it to the group.	
Study Goals	1. Acquire in-depth knowledge about the physics and image reconstruction underlying X-ray, CT, acoustical and magnetic resonance imaging; 2. Being able to solve elementary problems related to the theory mentioned in 1; 3. Being able to solve more advanced problems addressing the theory mentioned in 1 by combining mathematical skills and physical insight; 4. Able to acquire new knowledge about clinical a medical imaging technique; 5. Being able to present newly acquired knowledge about a medical imaging technique.	
Education Method	Lectures and an assignment.	
Literature and Study Materials	Book: Medical Imaging Signals & Systems, Jerry L. Prince, Jonathan Links. Upper Saddle River NJ: Prentice Hall, second edition, 2014, 544 pp, ISBN-13: 978-0132145183. We will provide Chapter 1 "Brief introduction to MRI physics" of the book: "Magnetic Resonance Image Reconstruction Theory, Methods, and Applications", 1st Edition - November 4, 2022, Editors: Mehmet Akcakaya, Mariya Ivanova Doneva, Claudia Prieto, Paperback ISBN: 9780128227268 , eBook ISBN: 9780128227466	
Assessment	Additional handouts wherever necessary. The course on Medical Imaging comes with an assignment for which you will need to hand in a paper and make a presentation. This is an obligatory part, for which you will get a grade from 1-10 (A). Additionally, there will be an exam also graded from 1-10 (B). The grades for both parts needs to be at least 6.0 in order to pass. The final grade is calculated by: $(A + B)/2$. It is not possible to take part in the exam if you have not made the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Pocket calculator.	
Elective	Yes	

AP3371	Radiological Health Physics	6
Responsible Instructor	D.S. Dihal	
Instructor	M. Schouwenburg	
Instructor	J.J. van Kammen	
Instructor	Dr. K.R. Huitema	
Contact Hours / Week x/x/x/x	0/0/8/8 (friday) = Note that the course starts earlier and ends later than Q3.	
Education Period	3 4	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Expected prior knowledge		
Summary	The course provides a broad overview of modern health physics. All subjects of health physics will be discussed in depth during a series of lectures. The course also consists of a practical part which is mandatory. The final exam is a theoretic exam in which both the student's ready knowledge and problem solving skills in the field of radiation protection are tested. This course brings participants up to the level of allround radiation protection experts, who may function as a coordination expert in a facility. With sufficient results on the final exam you receive an additional nationally recognized diploma "Coordinating Expert in Radiation Protection". The diploma is required for coordinating radiation experts, but also experts in radionuclide laboratories, medical clinical physicists and doctors of nuclear medicine need the diploma for their work.	
Course Contents	Lectures and tutorials on: - Radiation, radioactivity, decay. - Radiation sources. - Interaction of radiation with matter. - Methods of radiation detection. - Radiation dosimetry. - Radiation shielding. - Biological effects of ionizing radiation. - Internal dosimetry. - Natural and man-made sources/background radiation. - Radiation protection philosophy (protection principles). - Rules and regulations; organizational, procedural and administrative tasks. - Safety measures; operational radiation protection. - Radiation protection when handling open sources. Mandatory parts: - Practical experiments. These include prior preparation at home, active participation during the experiments, documenting results, reporting including discussion and conclusion/giving advice. All these aspect of the practical work will be monitored and scored during the course. - Several written assignments during the course. - Special modules on: Laser Safety, Communication, Organization & Task and Responsibilities, Risk Analysis and Ethics.	
Study Goals	This course will teach you how to work safely with sources of ionizing radiation. It is a broad introduction into the field of radiation protection and the application of ionizing radiation in science, medicine and industry. Because of this broad coverage of the subject it consists of theoretical parts, a mandatory practical part and special modules on topics like laser safety, communication, organisational aspects and risk analyses.	
Education Method	Oral lectures (classes), tutorials/instructions, assignments, lab experiments	
Literature and Study Materials	Massiot P, Jimonet C, European Radiation Protection Course: Basics. EDP Science (2014), ISBN: 978-2-7598-0703-1. Can be purchased on day 1 or 2 of the course.	
Practical Guide	The practical workbook will be made available on Brightspace on the second day of the course.	
Assessment	National exam in May. Resit in December. Two parts: - part 1: Multiple Choice; written closed book exam (max 33 points) - part 2: 4 problems; essay type questions, written open book exam (max 67 points)	
Permitted Materials during Tests	Part 1 (Multiple Choice part): non-programmable calculator and 'clean' dictionary. Part 2 (Essay type questions): All course materials like the textbook, notes, hand outs, etc.	
Special Information	For students from other universities: registration as a guest student is required for Brightspace access and registration of grades! See: https://www.tudelft.nl/en/student/administration/enrolment/enrolling-as-a-minor-student-or-guest-student/	
Elective	Yes	
Studyload/Week	1 day/week classes + 0.5 day/week labs + 0.5 day/week preparation	
Schedule	Note that the course starts earlier and ends later than Q3.	
Location	The detailed course program will be published on Brightspace in the second week of January. Reactor Institute Delft (building 50) Mekelweg 15 NL - 2629 JB DELFT	

AP3582	Medical Physics of Photon and Proton Therapy	6
Responsible Instructor	Dr. Z. Perko	
Instructor	Dr.ir. D. Lathouwers	
Instructor	Prof.dr. M.S. Hoogeman	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Some knowledge in basic programming with Python is required. The assignments can be made in any program that you are familiar with, but the big assignment notebooks are provided in only Python.	
Course Contents	The course deals with various topics concerning radiotherapy (mainly photon and proton): Introduction to radiotherapy Clinical aspects Radiobiology Biophysical models Proton therapy physics Dose shaping and plan optimization Dose calculation using clinical and Monte Carlo schemes Theory of dosimetry and in-vivo dosimetry Brachytherapy	
Study Goals	Main learning objectives 1. The student can define and explain the main clinical aspects of radiotherapy for different treatment modalities, list and describe the steps of a typical radiotherapy workflow. 2. The student can list and describe the basic principles of radiobiology and discuss their practical consequences for radiotherapy. 3. The student can recall and explain the function of biophysical models in radiotherapy, is able to apply these models and evaluate different treatment approaches by interpreting results. 4. The student can describe the main parts of the machinery used in various modes of radiotherapy and explain their role in administering dose. 5. The student can explain the clinical treatment planning process, can apply dose optimization methods and solve simplified planning problems, and can create and evaluate different treatment plans for idealistic model geometries. 6. The student can identify possible sources of errors arising in radiotherapy, and explain and describe correction methods used in the planning process and in practice for high precision radiotherapy. 7. The student can describe and explain the basic physics related to particle acceleration and beam transport, and can perform dose distribution calculations.	
Education Method	Lectures	
Literature and Study Materials	Handouts of powerpoint presentations and questions	
Assessment	The final grade for the course is calculated according to the following formula: $\text{Final grade} = 0.4 * \text{Homework grade} + 0.6 * \text{Exam grade}.$ To earn a passing grade, Homework grade has to be at least 5.5, and Exam grade and Final grade at least 6.0. Homework grade calculation The homework assignments are mandatory. The homework grade is based on the results of 3 small and 3 big assignments, with small and big assignments making up 25% and 75% of the total points for the homework, respectively. Small assignments must be made individually. Big assignments can be made either individually, or in pairs of 2, but you can only pair up with the same person once (so you will have to work with 3 different students if you want to make each big assignment in a pair). Pairs will always receive the same grade. All homework assignments will be handled via Brightspace (BS), with well-defined electronic submission deadlines at 23:59:59 on various dates. Late submission of all homework assignments will be allowed, but is only eligible for partial grade according to the following formula: $\text{Final assignment grade} = \text{assignment grade} * (1 - 0.2 * \text{number of started days after deadline}).$ Examination is written, and resit exam can be oral (by appointment), depending on the number of students. The partial grades for the exam and the homework assignments are valid for 1 year, i.e., for the given and the next academic years.	
Permitted Materials during Tests	Non-graphical/non-programmable pocket calculator.	
Elective	Yes	
Tags	Physics	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME-MP Obligatory electives: Select at least 10 ECTS from the following list	
Introduction 1	In below mentioned list Machine Learning (CSE2510) is a homologation course. This means that only students with another bachelor background as EE or CSE are allowed to include this course in the list of master courses.

AP3531	Acoustical Imaging	6
Responsible Instructor	Dr. D. Maresca	
Instructor	Dr.ir. D.J. Verschuur	
Instructor	Dr.ir. M.D. Verweij	
Instructor	Dr.ing. P.L.M.J. van Neer	
Contact Hours / Week x/x/x/x	0/0/2/2	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Expected prior knowledge	B.Sc.-level knowledge on wave phenomena and wave theory B.Sc.-level knowledge on Fourier transforms and signal processing. Basic knowledge of programming in Matlab or Python	
Course Contents	Applications of acoustical imaging in medical imaging, seismic imaging and non-destructive testing of construction materials. Review of Fourier and Radon transforms. The 3D acoustic wave equation and its properties. Derivation of the Kirchhoff and Rayleigh integral for acoustic waves. Forward and inverse wave field extrapolation in homogeneous and inhomogeneous media and its use in acoustical imaging. Resolution of the acoustic images and estimation of background propagation velocities. Medical ultrasound: basic physical principles, principles of ultrasound applications in medicine and neuroscience, Doppler imaging, ultrasound contrast agents. Ultrasound transducers and beamforming. The non-linear wave equation and harmonic imaging. Medical ultrasound imaging using contrast agents. Inverse scattering theory and full waveform inversion. Invited medical lectures on ultrasound imaging in cardiology and neurosurgery.	
Study Goals	Thorough understanding of the principles underlying acoustical imaging. Physical aspects and practical insights in the properties of wavefields in a medium. Understand the basic methodologies for geophysical and medical imaging, their resemblances and differences. Acquire comprehension of the potential and limits of current imaging techniques and the value of non-linear inversion techniques. Use theoretical concepts and bring them into practice.	
Education Method	Oral lectures and mandatory homework assignments in Matlab or Python.	
Literature and Study Materials	- Handouts of lecture notes - Book: The Principles of Quantitative Acoustical Imaging, by A. Gisolf and D.J. Verschuur, EAGE Publications BV (for sale at secretary office of section MI/CI, room F268)	
Assessment	Examination after successful completion of homework assignments. Oral exam if nr of students <20, otherwise written exam. For the homework assignments there is only pass/no-pass	
Remarks	The lectures are given by several lecturers: Dr. Eric Verschuur, Dr. Martin Verweij, Dr. David Maresca and Dr. Paul van Neer. Invited lecturers will cover specific topics, including medical lectures.	
Elective	Yes	
Tags	Image processing Matlab Physics	

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy).	
	Experience in programming and installing programming languages/libraries is required.	
	Without affinity/experience with python (and numpy) programming this course will become too time consuming as the project will become difficult.	
Course Contents	In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ - https://udlbook.github.io/udlbook/ Research papers will be made available through Brightspace.	
Assessment	Assignments are based on PyTorch: https://pytorch.org/ The final grade of the course consists of the following components: Presentation (weighting 10%). During the seminar a small group of students presents a paper. You will have to present once. Digital Exam (weighting 90%). Exam about the papers, lab assignments, and the theory (lectures). Project (Pass/Fail). In a small group of students you work on a deep learning paper reproducibility project. Final grade calculation: (pass project) * (0.1 * Presentation + 0.9 * Exam) A passing final grade for the course can only be earned when for all components at least a 5.0/pass is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentation: resit Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

IN4307	Medical Visualization	5
Responsible Instructor	T. Höllt	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic knowledge of linear algebra, calculus and programming is needed. The project will be implemented in MeVisLab and Python. Learning MeVisLab will be part of the practice assignments.	
Course Contents	Theory and practice (Notice project extends to Q2) of medical visualization. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measurement; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for medical data; advanced applications.	
Study Goals	By the end of the course, you should be able to LO1: Explain medical visualization algorithms and their applicability to medical problems. LO2: Discuss the advantages and disadvantage of medical visualization algorithms. LO3: Build a medical visualization system for a given problem: - Discuss a suitable visualization for a given medical problem. - Implement the most suitable solution. - Judge the performance of the implemented solution.	
Education Method	The course will be based on a combination of lectures and practical assignments. A final project will be developed in Q2	
Literature and Study Materials	Course slides, instructions for project and assignments, and selected literature via Brightspace. Chapters from the following book can be used as alternate view on the topics but are not mandatory: Visual Computing for Medicine 2nd Edition Theory, Algorithms, and Applications Authors: Bernhard Preim Charl Botha Morgan Kaufmann Book available in electronic form via the TUDelft library.	
Assessment	The final grade of the course consists of the following components: Written exam Q1 (weighting 40%) Final project Q2 (weighting 60%) The final project will be the design and implementation of a visualization system for a given medical problem. The final project will be carried out in teams of three. The deliverables for the final project will be a report (paper), the results (e.g., code) and a short video presenting the project (i.e. screencast). Final grade calculation: $0.4 * \text{Written exam} + 0.6 * \text{Final project}$ A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Q2 In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Final project: re-submit deliverable (the project resit is not automatic and must be requested within 2 weeks after publishing the grade). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	Notes and written material. No computers.	
Special Information	It is necessary that you register/enroll on Brightspace for this course. In the first lecture, details on the evaluation and practical information on the course will be given.	
Co-Instructor	Prof.dr. E. Eismann	

NB4020	High-Resolution Imaging	4
Responsible Instructor	Dr. A. Jakobi	
Instructor	Dr.ir. J.P. Hoogenboom	
Instructor	S. Stallinga	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Microscopy is an essential technique in the life science and developments in microscopy have almost always led to new biological discoveries. In the course High Resolution Imaging we introduce the state of the art techniques to observe cells, tissue, and biological molecules at the highest possible resolution. We provide a working knowledge of various high resolution imaging techniques, present the basic principles and limitations behind these techniques, and also discuss how to prepare biological samples for the different forms of microscopy. Examples from recent scientific literature will be used to illustrate what can be and has been achieved. Targeted homework, lab demonstrations, and experimental as well as theoretical exercises will help to familiarize you with the different types of microscopes and the data they provide. Specifically, the course includes lectures and demos on: Superresolution fluorescence microscopy. Scanning electron microscopy (SEM), Transmission electron microscopy (TEM), and Scanning transmission electron microscopy (STEM). Electron cryo-microscopy, and methods to calculate 3D structures from 2D projections. Correlative light and electron microscopy and techniques for large-scale and volume imaging.	
Study Goals	After completion of this course, the students should: 1. Understand the basic principles and limitations for focusing light and electron beams in microscopes. 2. Be able to explain the principles for contrast and image formation in the SEM, STEM, and TEM. 3. Understand the properties of fluorescent labels and the basic principles used in superresolution light microscopy 4. Understand the basic principles of 3D electron microscopy. 5. Be able to explain the role of sample preparation in high-resolution microscopy, understand the steps of fixation (chemical and cryogenic), staining, embedding, sectioning, and labelling, and reproduce the main considerations for each step. 6. Be able to estimate the limitations of each method and decide which of them would be suitable in a particular experiment. 7. Be able to judge the validity of a microscopy approach presented in an experimental work.	
Education Method	Lectures, homework, image processing exercises. If circumstances permit discussion groups, practicals and demonstrations may also be part of the course.	
Literature and Study Materials	Required: Lecture notes & articles provided during the course.	
Assessment	The final grade is comprised of the grades for the assignments and the exam grade where each individual grade has to be 5.5 or higher. The maximum weight for all assignments together is 30%. The grades for the assignments remain valid for the retake. The retake may be conducted as an oral exam if <5 students subscribe for the retake. There is no possibility to retake an assignment. Assignment grades expire at the end of the academic year. A student retaking the course in the next academic year has to do the assignments of the new course.	
Enrolment / Application	Registration for written exams is via Osiris and deadlines are determined by TU Delft regulations on exam registration. Enrollment in Brightspace is considered registration for any and all other assessments, students must enroll in Brightspace for the course no later than the first week of classes.	
Location	Delft	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME Master 2nd Year

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME Project / Electives - Select 1 out of 3: Internship, JIP or Elective Courses (15 ECTS)	
Introduction 1	In this area of 15 ECTS it is possible to choose for doing an Internship, the Joint Interdisciplinary Project or to select 15 ECTS elective courses which are mentioned under the 1st year of this master programme.

BM51015	BME Research Internship	15
Responsible Instructor	A.J. Loeve	
Instructor	Ir. M. Cuschieri	
Contact Hours / Week x/x/x/x	x/x/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Internship consists of three months (10-12 weeks) of research work outside TU Delft to get working experience in another environment than TU Delft. This could either be a commercial company, a medical center, a research center or another university within the Netherlands or abroad.	
Study Goals	After the Internship, the student should be able to: - recognize how an internship could contribute to their personal development. - define their own learning goals for personal development. - reflect critically on their work and their personal development. - reflect on how well the experienced working environment and work fits their skills, interests and personality.	
Education Method	The Internship is an integral part of the second MSc year. The Internship, the Literature study, and the final MSc project, preferably all concern the same field of research. This gives you a chance to really dive into the subject of your master thesis. Throughout their internship, students undertake a project task defined in consultation with the host institute and the TUD-BME supervisor. It is recommended that Dutch students undertake their internship abroad. Before they plan their internship abroad, it is encouraged that the students contact the International Office at 3mE (internationaloffice-3me@tudelft.nl) for further information. Internships are usually arranged via one of the staff members in the students chosen track. Students are encouraged to contact BME staff, preferably their (intended) thesis project advisor, at the start of the internship selection process to act as a TUD-BME supervisor. The TUD-BME master coordinator must give approval before the internship is started. Check the "Master Biomedical Engineering" Brightspace page and the Flowcharts and Guidance Doc 2A that you find there to know all about the required procedures and the location of any required forms. Furthermore, BME students, instead of undertaking an Internship, can also join an interdisciplinary team under the frame of the Joint Interdisciplinary Project (15 EC). Information about JIP can be found at the following website: https://www.jointinterdisciplinaryproject.nl. ===CHECK THE BME STUDY GUIDE, THE FLOWCHARTS AND GUIDANCE DOCS 2A AND 2D ON THE "MASTER BIOMEDICAL ENGINEERING" BRIGHTSPACE PAGE FOR MORE AND UP-TO-DATE INFORMATION===	
Practical Guide	We do not maintain a separate Brightspace page for this course. please find everything you need on the "Master Biomedical Engineering" Brightspace page and in the Study Guide, Flowchart and Guidance Docs provided there.	
Assessment	You are required to write a brief report about your work and your experiences during the Internship. Some students write it in the form of a scientific paper. It will contain: 1) Internship report - About the contents 2) Internship reflection - About your learning goals You will have to arrange getting your work and reflection assessed and signed off according to the procedures set out in the Master Biomedical Engineering (M-BME) Year 2 Flowchart and GuidanceDoc A. You can find these on the M-BME Brightspace page. Also check those for details and for where to find all necessary forms and procedures. ===CHECK THE BME STUDY GUIDE, THE FLOWCHARTS AND GUIDANCE DOCS A AND D ON THE "MASTER BIOMEDICAL ENGINEERING" BRIGHTSPACE PAGE FOR MORE AND UP-TO-DATE INFORMATION===	
Department	3mE Department Biomechanical Engineering	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Biomedical Engineering

BME Graduation Project Obligatory (45 ECTS)

BM51010	BME Literature Research	10
Responsible Instructor	A.J. Loeve	
Instructor	Ir. M. Cuschieri	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Literature study prepares you for your MSc thesis. One of the reason to start with a literature study is that you quickly achieve an overview of your research topic and you will be prevented to start studying a topic or designing a device that already exists.	
Study Goals	The goal of the Literature study is twofold: Firstly, you will demonstrate your capability to: - systematically collect and interpret large quantities of information - identify the most interesting and relevant questions or problems in the field of your MSc thesis. - summarize the information in a systematic overview representing the "State of the Art" in the field of your MSc thesis Secondly, you will write a literature report in which you have to: - perform an in-depth discussion, in which you demonstrate your ability to critically evaluate hypotheses and results - demonstrate a sound line of reasoning - draw conclusions that point towards new research opportunities - demonstrate a writing style and quality that equals the required level of your MSc thesis	
Education Method	The Literature study is individual work. The student and the TUD-BME thesis advisor together decide on the topic. The student is in charge of the planning. Although you are in charge, this doesn't mean that you have to do this in isolation. Ask your thesis advisor for advice, but also contact a librarian or an expert in the field for advice in on your search for literature. Writing on a scientific level is difficult, so the feedback that you receive on the drafts of your literature report will be very valuable. The literature search is typically divided in phases. You can find more information on how to approach your literature study in the Guidance Docs that we provide on the "Master Biomedical Engineering" Brightspace page: - Guidance Doc 2B - Literature review - Guidance Doc 2D - Writing reports Please mind that we do not maintain a Brightspace page for this course. All information is provided via the "Master Biomedical Engineering" Brightspace page.	
Practical Guide	We do not maintain a separate Brightspace page for this course. please find everything you need on the "Master Biomedical Engineering" Brightspace page and the Flowcharts and Guidance Docs provided there.	
Assessment	Report	
Department	3mE Department Biomechanical Engineering	

BM51035	BME MSc Thesis	35
Responsible Instructor	A.J. Loeve	
Instructor	Ir. M. Cuschieri	
Contact Hours / Week x/x/x/x	0/0/x/x	
Education Period	None (Self Study)	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The Master thesis project is the final part of the BME program. Ideally, the project is undertaken in collaboration with a clinical partner (e.g. LUMC, Erasmus MC, VUMC, UMCU, Amsterdam UMC, RdGG, UMCG or MUMC). The student, together with the TUD-BME thesis advisor and the clinical advisor, will define the research project. The student then prepares the MSc thesis as a project report. The thesis work is evaluated by way of an oral presentation (graduation colloquium) by the candidate and an oral examination in the form of a defence with an MSc examination committee composed according to the rules set out in Form 6 (https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/msc-forms). The examination committee may also include external examiners from research institutes or industrial partners. If a student undertakes the graduation project at a company, a graduation agreement must be signed: use the template of the TUD -3mE graduation agreement. Check the "Master Biomedical Engineering" Brightspace page for further information and for links to the appropriate forms. All BME students must fill out a BME graduation agreement form at the start of their graduation period in their second year. The template of this form is available on the Brightspace of the BME Master and should be sent to the appropriate people. See the Flowcharts and Guidance Docs 2C and 2D on the BME Master Brightspace for further details.	
Study Goals	The overall objective of this assignment is to demonstrate a sufficient academic level in the field of BioMedical Engineering.	
Education Method	Self study with regular supervision.	
Practical Guide	We do not maintain a separate Brightspace page for this course. please find everything you need on the "Master Biomedical Engineering" Brightspace page and the Flowcharts and Guidance Docs provided there.	
Reader	We do not maintain a separate Brightspace page for this course. please find everything you need on the "Master Biomedical Engineering" Brightspace page and the Flowcharts and Guidance Docs provided there.	
Assessment	MSc Thesis, presentation and defence.	
Department	3mE Department Biomechanical Engineering	

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